Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

There are 3,041,714 entries in the January 2023 parquet file. Assuming that the rest of the files are of similar sizes, it may not be computationally feasible to do the EDA on the free version of colab notebook because of limited resources.

1.1.1. Sample the data and combine the files

To handle the file size, we can work with a sample of the data. For all the days in a month, we can iterate through the hours and select 5% values randomly from those.

After sampling the data, there are a total of 1,896,400 entries for us to work with.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

The index is consistent with values ranging from 0 to 1,896,400. Also dropped helper columns – date and hour.

2.1.2. Combine the two airport_fee columns

The are 2 columns with the same name – airport_fee and Airport_fee. By checking the distribution of values – airport_fee has about 92% of values as null whereas, Airport_fee has about 11% null values. Considering this, both these columns can be consolidated by taking Airport_fee as the primary column, and combining them using the combine_first function in pandas. The resultant column is renamed as “airport_fee”, then column Airport_fee is dropped as it is redundant.

Negative fare values in column fare_amount: 0

Negative fare values in column extra: 3

Negative fare values in column mta_tax: 73

Negative fare values in column tip_amount: 0

Negative fare values in column tolls_amount: 0

Negative fare values in column improvement_surcharge: 78

Negative fare values in column total_amount: 78

Negative fare values in column congestion_surcharge: 56

Negative fare values in column airport_fee: 15

Most negative fares have the RatecodeID of either 1 or 2. Since there are a very few rows having negative monetary values, and assuming that negative monetary values indicate refunds or trip cancellations – we can drop these rows.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

The columns 'passenger_count', 'RatecodeID', 'store_and_fwd_flag', 'congestion_surcharge', 'airport_fee' have about 3.5% of values as nulls. Rest of the columns do not have null values.

2.2.2. Handling missing values in passenger_count

A driver cannot make a trip with 0 passengers. Hence this value is most likely to be an error/invalid value. After converting the rides with 0 passenger_count to null – the total number of nulls in this column rises to about 5%. Since the majority of the rides has a passenger_count of 1 (~72%) the missing values in this column can be imputed by mode, i.e., 1.

2.2.3. Handle missing values in RatecodeID

The ratecodeid has about 3.4% missing values, and also includes an invalid category of 99 with consists of about 0.5% of the total. These values can be imputed with mode, i.e., 1 since 91% of the trips belong to this category.

2.2.4. Impute NaN in congestion_surcharge

About 89% of the trips face a congestion surcharge of 2.5. The missing values in this column can be imputed with mode, i.e., 2.5.

2.2.5. Handling missing values in airport_fee and store_and_fwd_flag

About 88% of the rides has zero airport fees, hence the missing values in this column can be replaced by the mode, i.e., 0. About 96% of the rides has the flag N for store_and_fwd_flag, hence the missing values in this column can be replaced by the mode, i.e., N.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Entries where trip_distance is nearly 0 and fare_amount is more than 300 is likely due to distance recording error or GPS failure. These rows can be dropped because we cannot infer the true distance. Entries where trip_distance = 0 and fare_amount = 0 but pickup is different from dropoff zone is likely an error because If pickup and dropoff zones differ, distance cannot be zero. Trips having trip_distance > 250 miles is likely an error / outlier since the taxi service serve journeys within the city and can be dropped. For entries with payment type as zero can be replaced with 5 (unknown) since 0 is not defined in the dictionary. Additionally, a new feature is speed_mph is derived which gives the speed of the taxi in mph. Some unrealistic values are removed assuming that they are outliers. Outliers beyond 99th percentile for the columns “trip_distance", "fare_amount”, "tip_amount", "tolls_amount", "total_amount" and “trip_duration" are dropped.

2.3.2. Standardising values

The values in the column’s vendorid, payment_type and RatecodeID can be standardised by mapping its encoded value with its corresponding text value. This increases readability when visualizing the data.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

VendorID: Categorical

tpep_pickup_datetime: Numerical (Temporal)

tpep_dropoff_datetime: Numerical (Temporal)

passenger_count: Categorical (Ordinal)

trip_distance: Numerical

RatecodeID: Categorical

PULocationID: Categorical

DOLocationID: Categorical

payment_type: Categorical

pickup_hour: Categorical

trip_duration: Numerical
fare_amount: Numerical

extra: Numerical

mta_tax: Numerical

tip_amount: Numerical

tolls_amount: Numerical

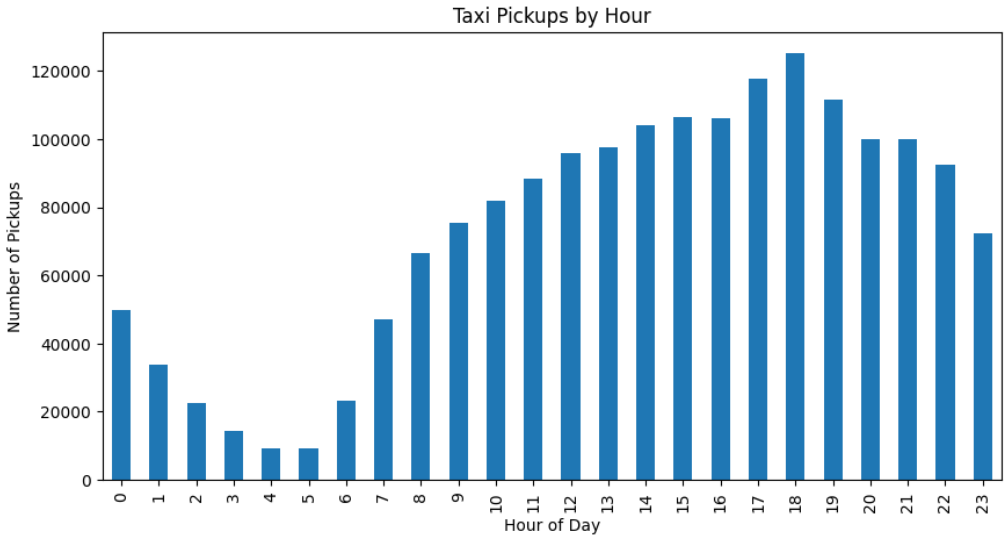
improvement_surcharge: Numerical

total_amount: Numerical

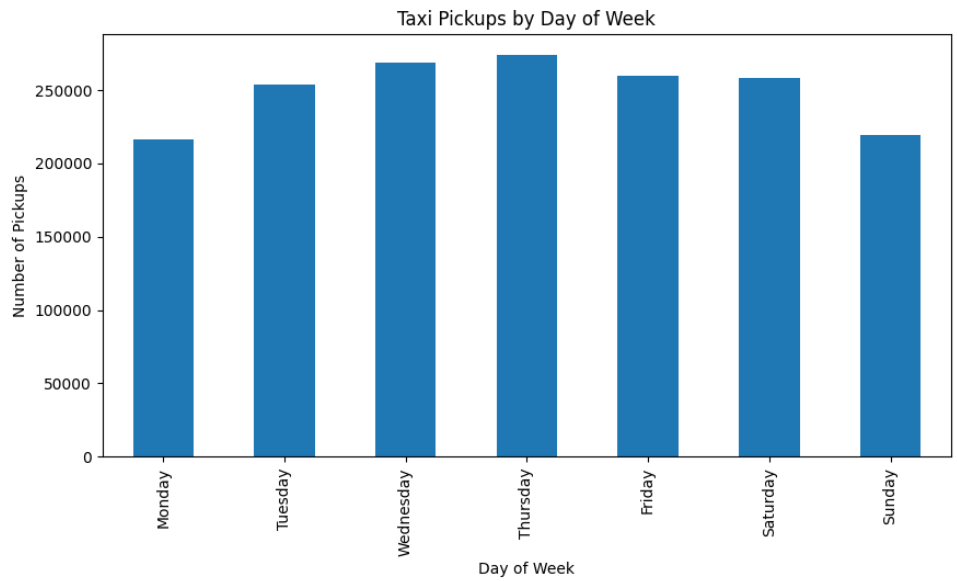
congestion_surcharge: Numerical

airport_fee: Numerical

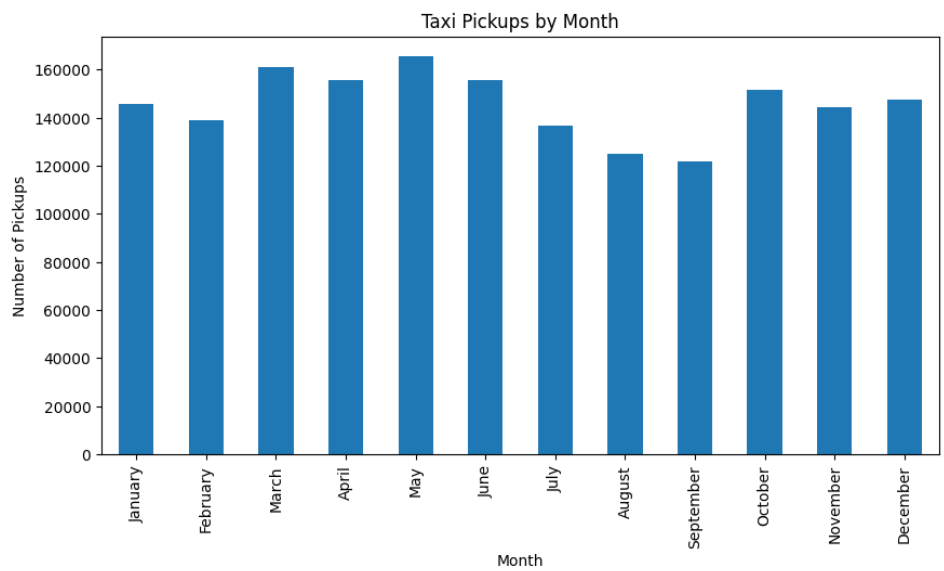
3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months



Taxi demand is lowest between 2 AM – 5 AM. Pickups start increasing after 6 AM, indicating the morning commute. Demand steadily rises through the day. The peak occurs around 5–6 PM, corresponding to the evening rush hour. Demand gradually decreases after 7 PM, though late evening still has relatively high activity.



The number of pickup counts increase from Tuesday to Thursday, with Thursday being the highest. The demand remains high on Friday and Saturday. Sunday shows a noticeable drop which continues till Monday.



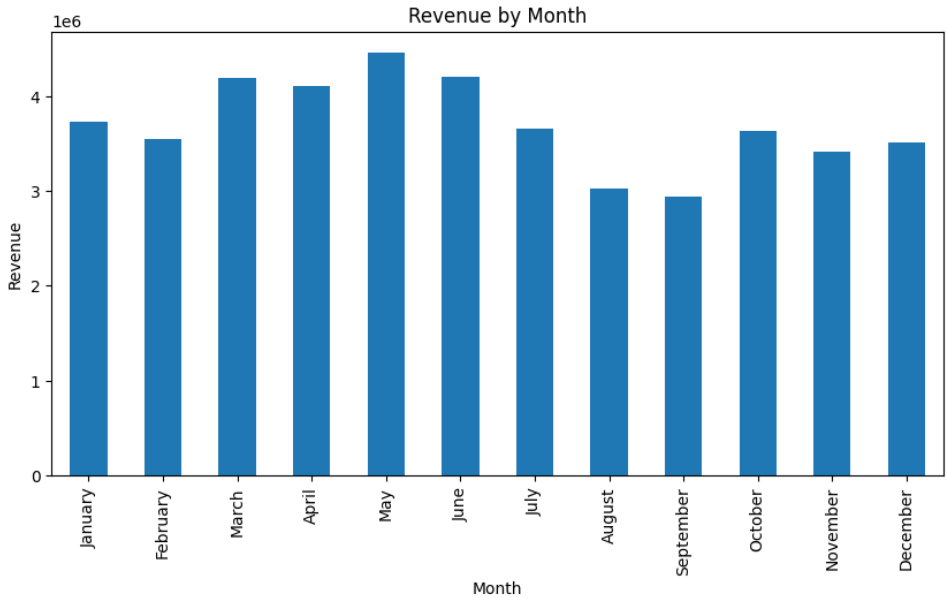
Pickup demand fluctuates moderately throughout the year. The month of May shows the highest demand. The period between July – September show relatively low pickup counts, whereas the period between March – May and October – December have the high pickup count.

3.1.3. Filter out the zero/negative values in fares, distance and tips

A separate dataframe which consists of fare amount, trip distance and total amount > 0 is created with the assumption that trips having 0 values

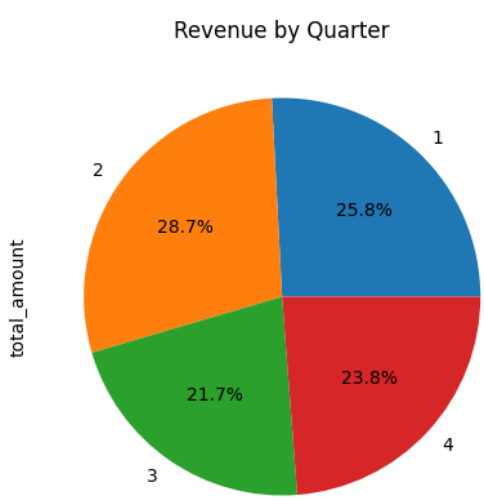
in these columns are invalid. Also, a separate dataframe is created for tip amount > 0 since cash tips are not recorded. Having separate dataframes helps us analyse valid trips separately, and since cash tip is not recorded, high number of zero tip amount is biased towards tips completed via cash payments.

3.1.4. Analyse the monthly revenue trends



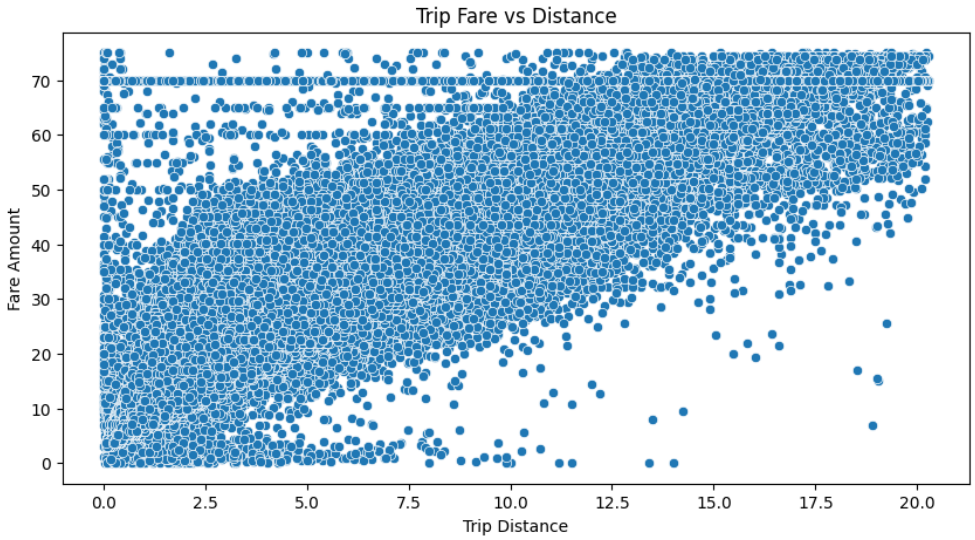
Revenue fluctuates throughout the year. There is a noticeable increase in revenue from February – May, after which it decreases from June - September. August - September show the lowest revenues, and the period between March – June has the highest revenue.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

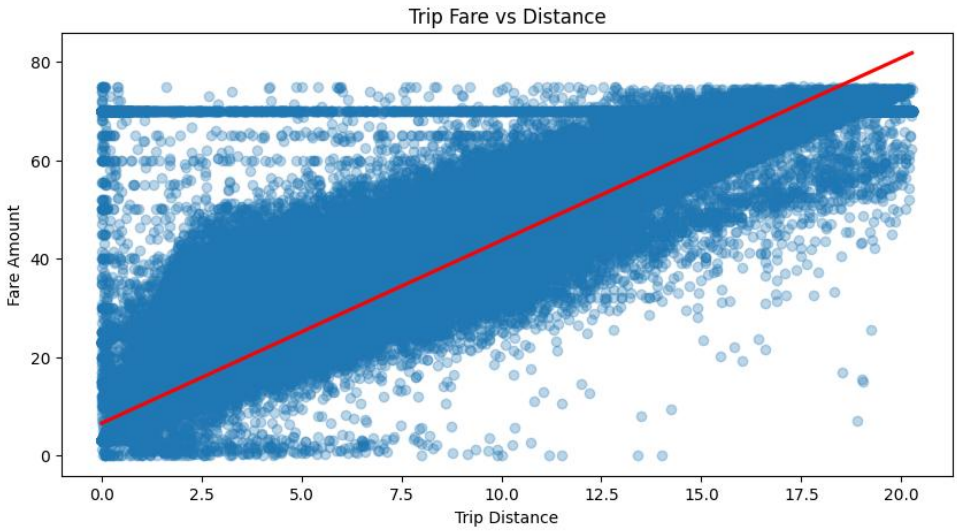


Q2 contributes the largest share of revenue at 28.7%. Q3 contributes the least at 21.7%.

3.1.6. Analyse and visualise the relationship between distance and fare amount

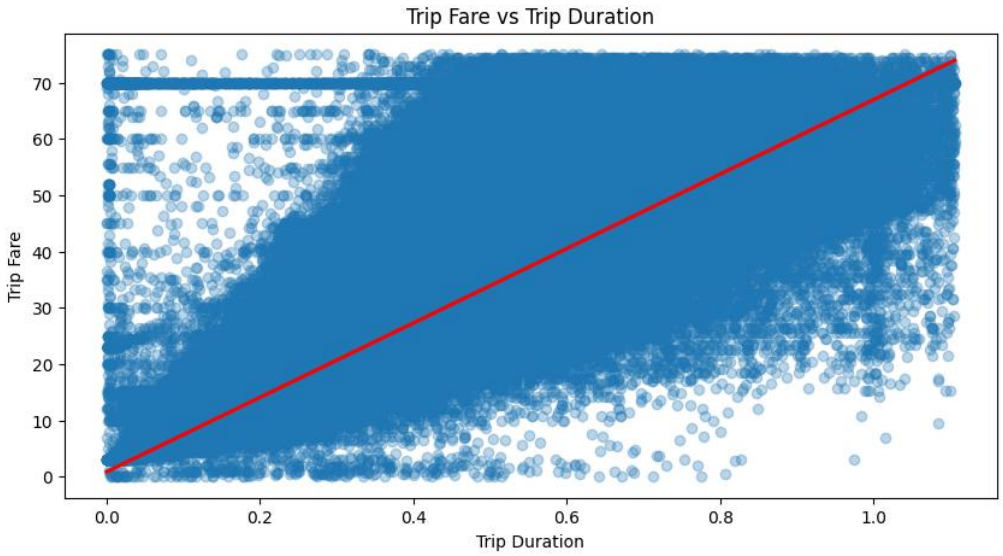


There is a positive relationship between distance and fare. There is a clear upward trend, indicating that as trip distance increases, fare amount also increases.

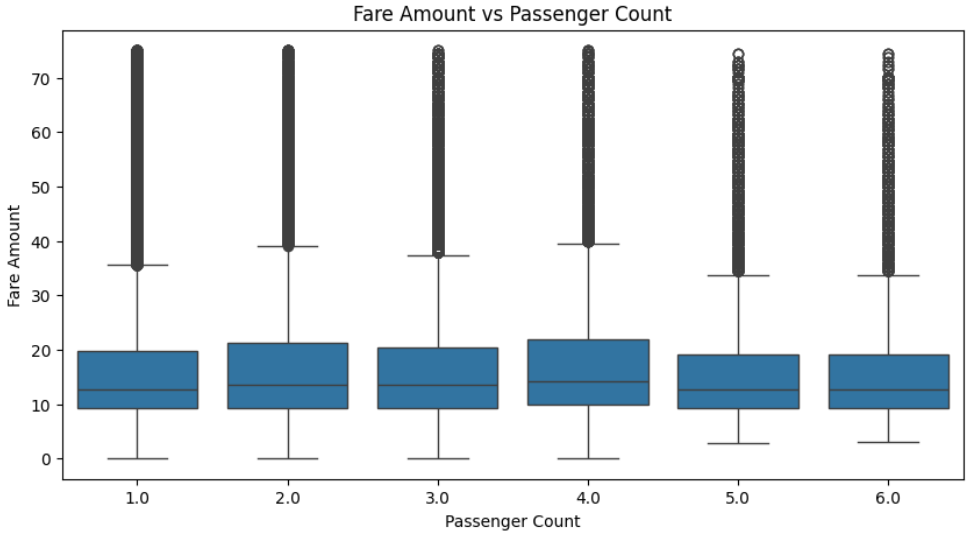


Correlation between trip distance and fare amount is: 0.9490.

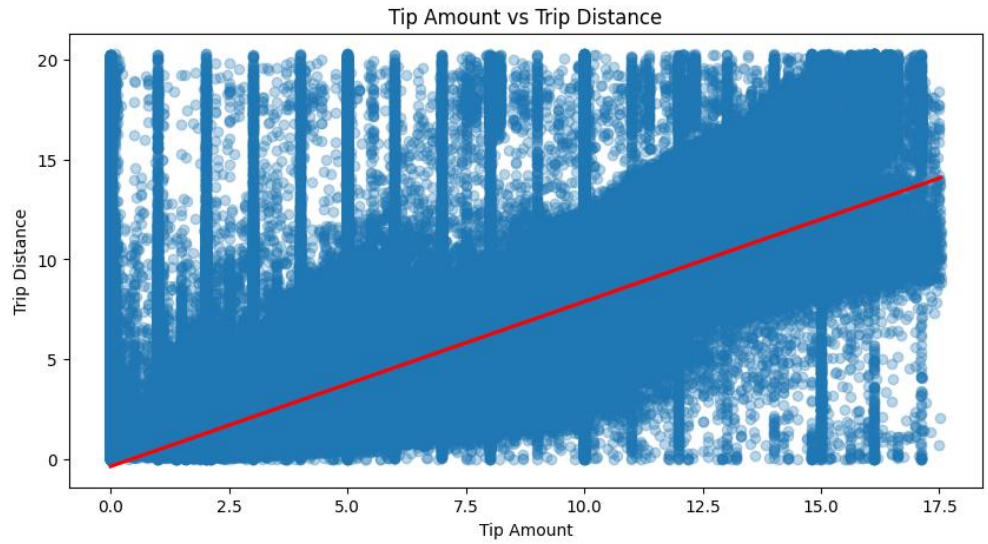
3.1.7. Analyse the relationship between fare/tips and trips/passengers



The plot shows a clear positive relationship between trip duration and fare. As trip duration increases, fare amount also increases, which aligns with taxi pricing (base fare + time + distance). The regression line confirms a strong upward trend. There is greater variability in fare for longer trips. Correlation between trip duration and fare amount is: 0.8786

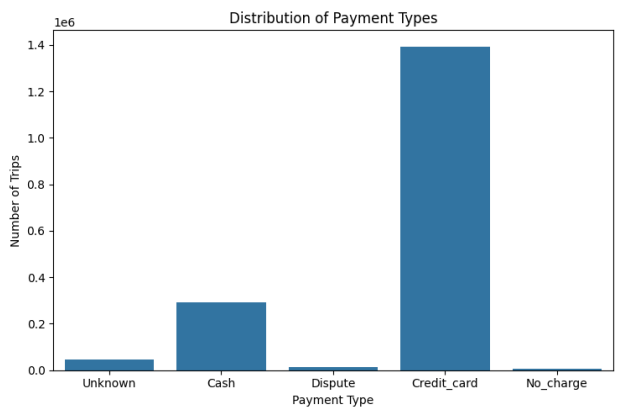


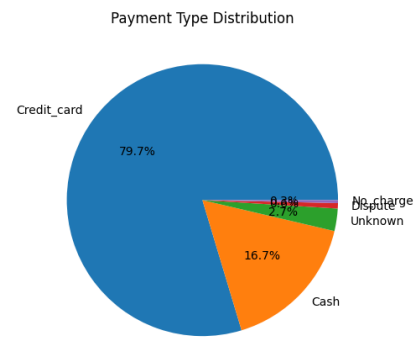
Median fares are very similar across passenger counts (1–6 passengers). Fare distributions overlap heavily between categories. Passenger count does not significantly affect fare amount.



There is a positive relationship between trip distance and tip amount. Longer trips generally receive higher tips. Correlation between trip distance and tip amount is: 0.7205

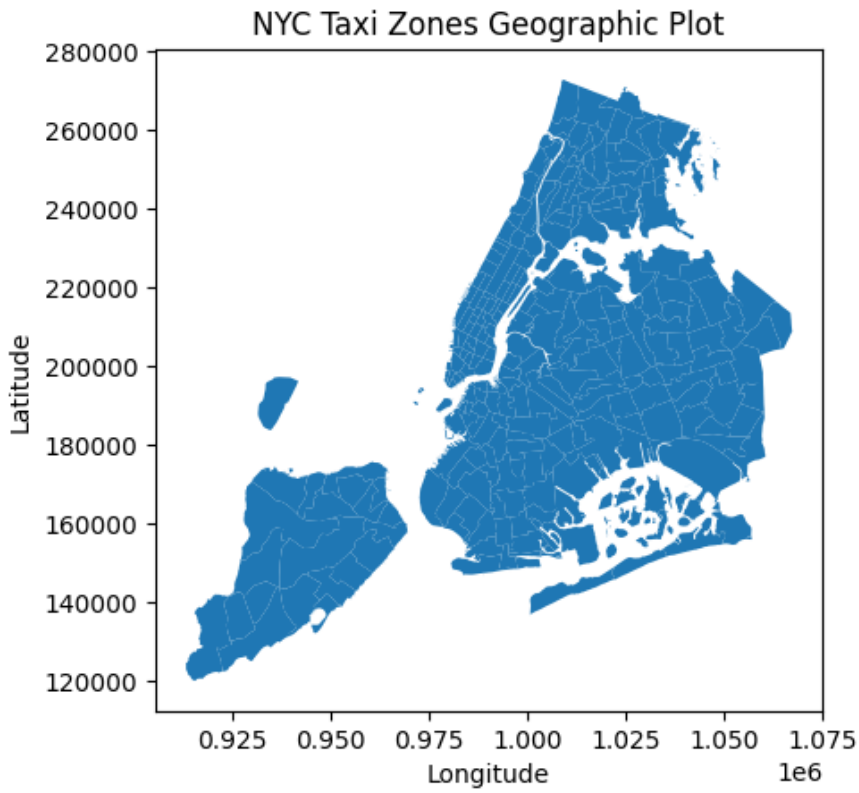
3.1.8. Analyse the distribution of different payment types





The vast majority of trips are paid using credit cards at ~80%. This suggests that digital payment methods are the most preferred option for taxi passengers. Cash is the Second Most Common Method at ~17%. Cash payments represent a significantly smaller share compared to credit cards. However, they still account for a notable portion of transactions, indicating that some passengers still prefer traditional payment methods. Other Payment Types are Rare.

3.1.9. Load the taxi zones shapefile and display it



The geographic plot illustrates how NYC is divided into taxi zones to capture spatial patterns of taxi usage. The denser zoning in Manhattan highlights its role as the primary hub of taxi activity, while larger zones in outer boroughs reflect relatively lower trip density.

3.1.10. Merge the zone data with trips data

The pd.merge() function combines two DataFrames: df_valid and zones. The merge is performed using: PULocationID from the taxi dataset and LocationID from the zones dataset. It is merged using left join, which means all rows from the taxi dataset (df_valid) are preserved and matching zone information is added when available, if no match exists, the new columns will contain NaN values. This merge enriches the taxi dataset by adding geographical context to each pickup location**.**

3.1.11. Find the number of trips for each zone/location ID

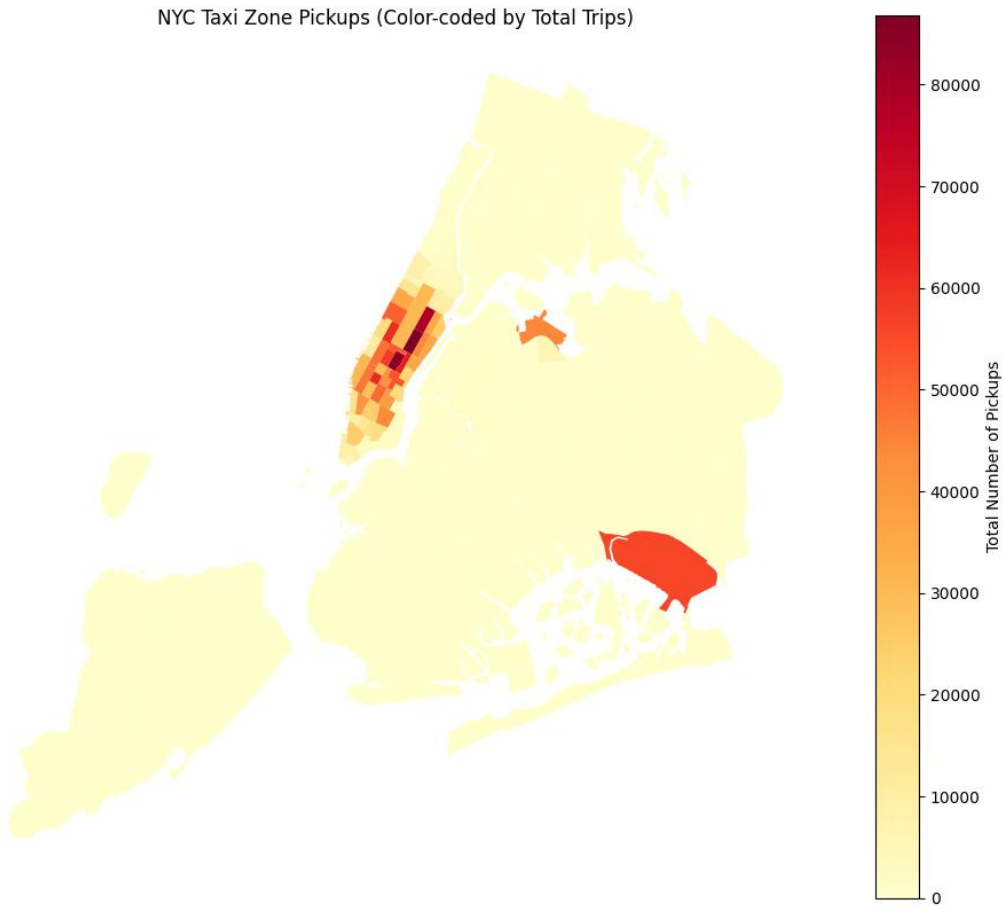
The groupby('PULocationID') groups the dataset based on the pickup location ID. The size() counts the number of trips originating from each pickup location. The reset_index(name='total_trips') converts the result into a DataFrame and labels the count column as total_trips. This allows us to identify which taxi zones have the highest pickup demand.

3.1.12. Add the number of trips for each zone to the zones dataframe

The zones dataframe is merged with the trip counts dataset. The merge

matches LocationID from the zones dataset and PULocationID from the trip count dataset. A left join ensures that all taxi zones remain in the dataset, even if no trips occurred there. Zones with no recorded pickups will have NaN values. These are replaced with 0, indicating no trips occurred in those zones.

3.1.13. Plot a map of the zones showing number of trips



Manhattan is the Primary Taxi Demand Hub. The darkest zones are concentrated in Manhattan, particularly Midtown and Lower Manhattan. These areas show the highest number of taxi pickups, indicating they are the main centers of taxi activity. Airports are Major Pickup Locations. A prominent high-demand zone appears in Queens, corresponding to LaGuardia Airport. Outer Boroughs Show Lower Taxi Activity. Brooklyn, Queens (non-airport areas), Bronx, and Staten Island appear mostly in lighter colors. This indicates significantly fewer taxi pickups compared to Manhattan.

3.1.14. Conclude with results

Hourly Trends

Taxi demand is lowest during early morning hours (2 AM – 5 AM). Pickups begin increasing after 6 AM, reflecting the morning commute. Demand peaks around 5 PM – 6 PM, corresponding to the evening rush hour. Activity remains relatively high during the evening and late-night hours.

Weekly Trends

Pickup demand gradually increases from Monday to Thursday. Thursday, Friday, and Saturday show the highest taxi activity, driven by commuting, nightlife, and social travel. Sunday sees a noticeable decline compared to the rest of the week.

Monthly Trends

Taxi activity is fairly consistent throughout the year. The period from March – June show highest pickup volumes. The period from July – September show lowest pickup volumes.

Trends in Revenue Collected

Monthly taxi revenue varies between approximately $3M and $5M. Revenue tends to increase during months with higher trip demand. Spring months appear to generate slightly higher revenues than winter and summer months.

Quarterly Revenue Trends

Revenue varies slightly by quarter, with Q2 having the highest contribution at 28.7% followed by Q1 at 25.8%. Q3 has the lowest contribution at 21.7% followed by Q4 AT 23.8%.

Fare vs Trip Distance

A strong positive relationship exists. Longer distances produce higher fares, as expected from the meter pricing model.

Fare vs Trip Duration

Fare increases with trip duration. Longer trips result in higher fare amounts, especially in heavy traffic conditions.

Fare vs Passenger Count

edian fares are very similar across passenger counts (1–6). Passenger count does not significantly affect fare.

Tip Amount vs Trip Distance

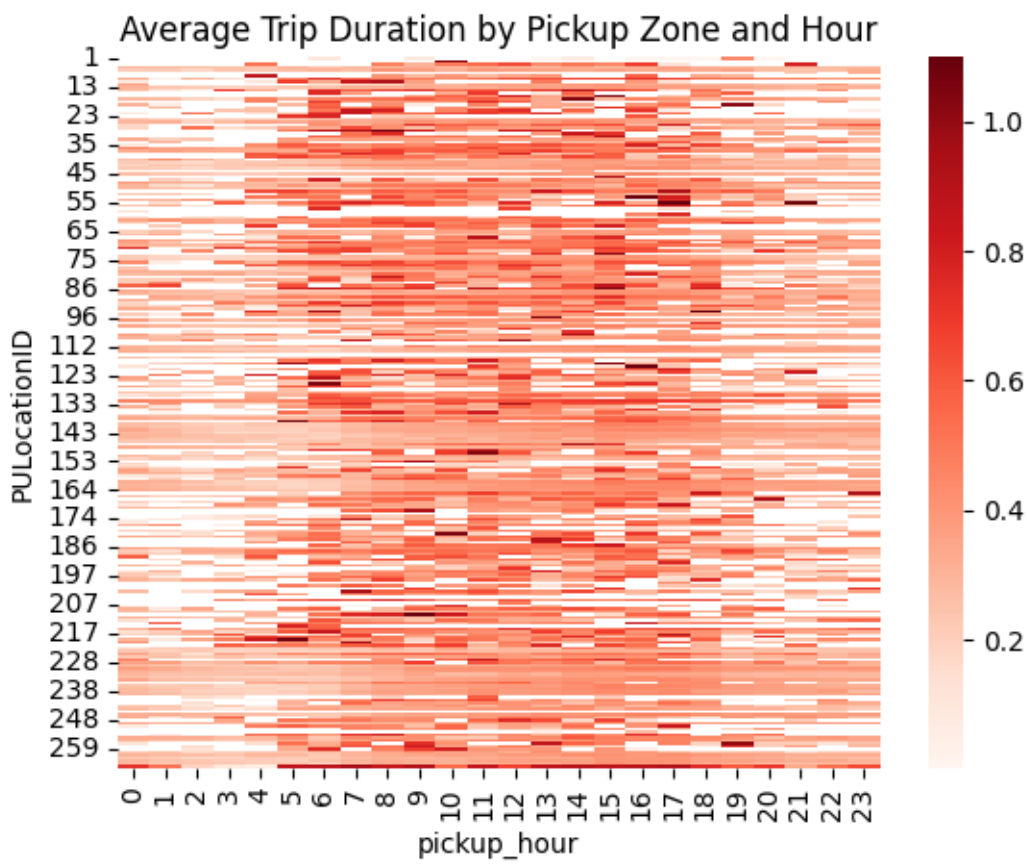
Tips generally increase with trip distance. However, the relationship shows high variability. Longer trips tend to produce higher tips, but tipping behavior is less predictable than fare.

Busiest Taxi Zones

Manhattan shows the highest pickup density, particularly Midtown and Lower Manhattan. Airport zones (LaGuardia) also show high pickup counts. Outer boroughs show significantly lower pickup density.

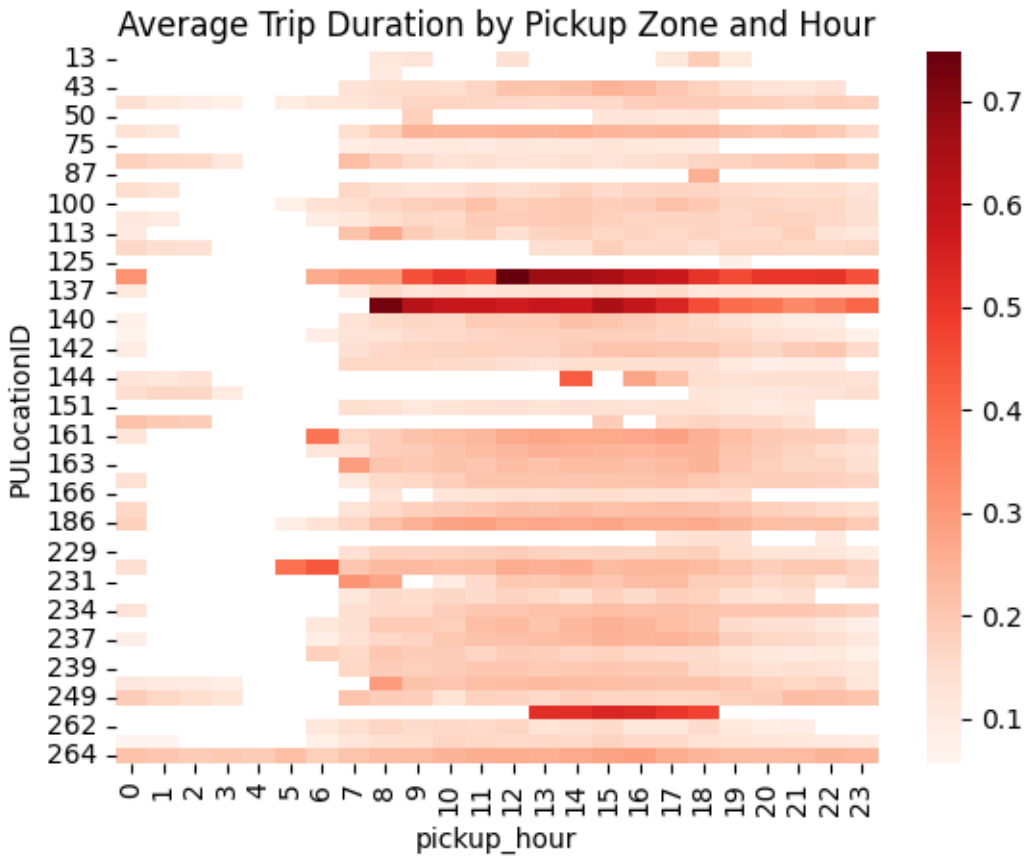
3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes



Across most pickup zones, trip durations tend to increase during daytime hours from 6 AM to 7 PM. These hours show darker shades, indicating longer average trip durations, likely due to traffic congestion. Late night and early morning hours (7 pM – 5 AM) generally show lighter colors, meaning shorter travel times because traffic levels are low. The unfiltered heatmap appears noisy because routes with very few trips can produce

misleading averages.



After filtering for routes with ≥ 50 trips, the heatmap becomes clearer and more structured and true patterns of traffic congestion and demand become visible. These results are statistically more reliable. Some pickup zones display consistently darker bands across many hours. This suggests that these zones frequently experience longer travel times and may correspond to high-traffic areas such as Manhattan or airport routes. These zones represent potential congestion hotspots.

How Identifying High-Traffic / High-Demand Routes Helps

1. Better Taxi Fleet Allocation

Taxi operators can position more vehicles in high-demand zones during peak hours, reducing passenger wait times.

2. Improved Traffic Management

City planners can use these insights to identify traffic bottlenecks, improve road infrastructure and adjust traffic signal timing.

3. Surge Pricing and Demand Forecasting

Ride-hailing services can adjust pricing dynamically and forecast future demand patterns

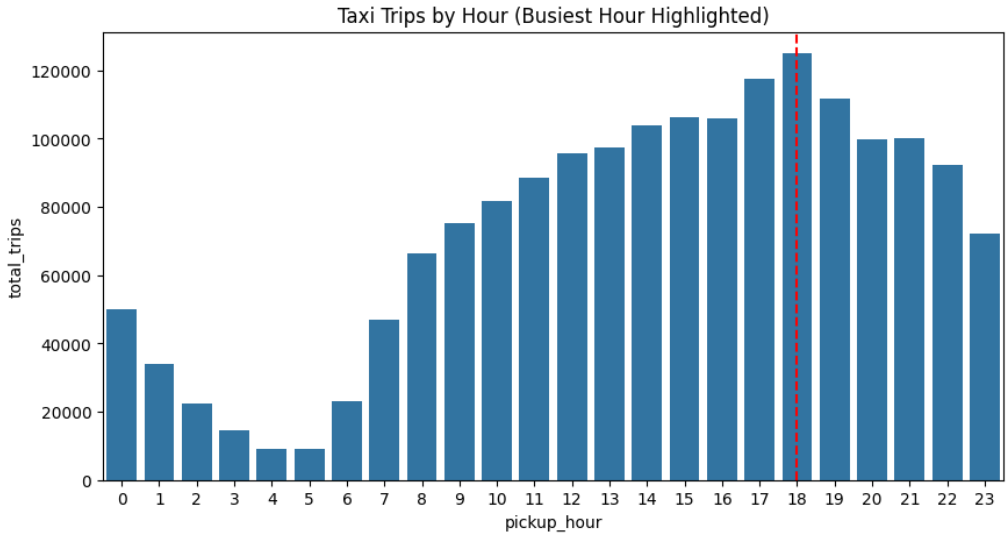
4. Route Optimization

Drivers and navigation systems can avoid congested routes and suggest faster alternatives

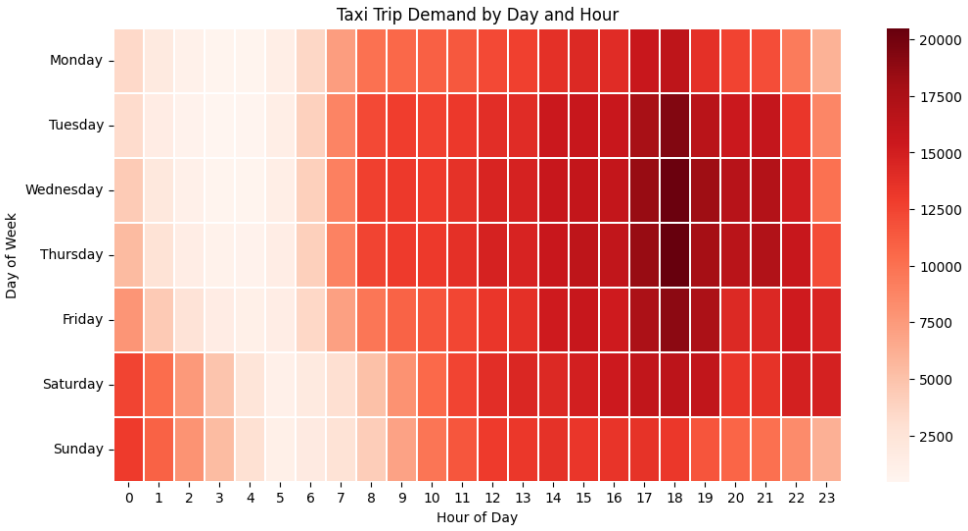
5. Urban Mobility Planning

Understanding these patterns helps planners evaluate public transport coverage and areas underserved by transit and mobility demand across neighborhoods

3.2.2. Calculate the hourly number of trips and identify the busy hours



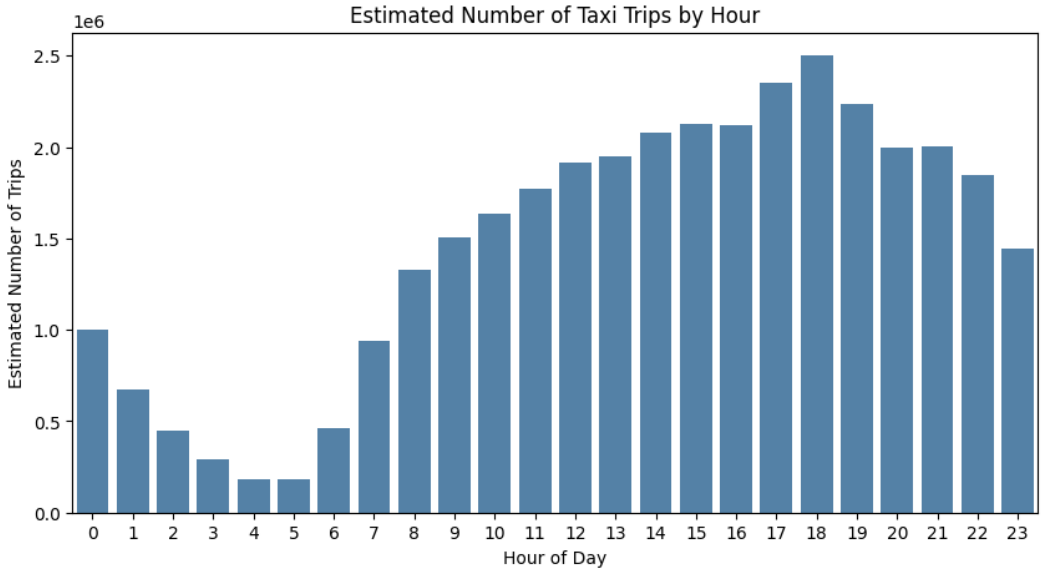
6 PM is the busiest hour.

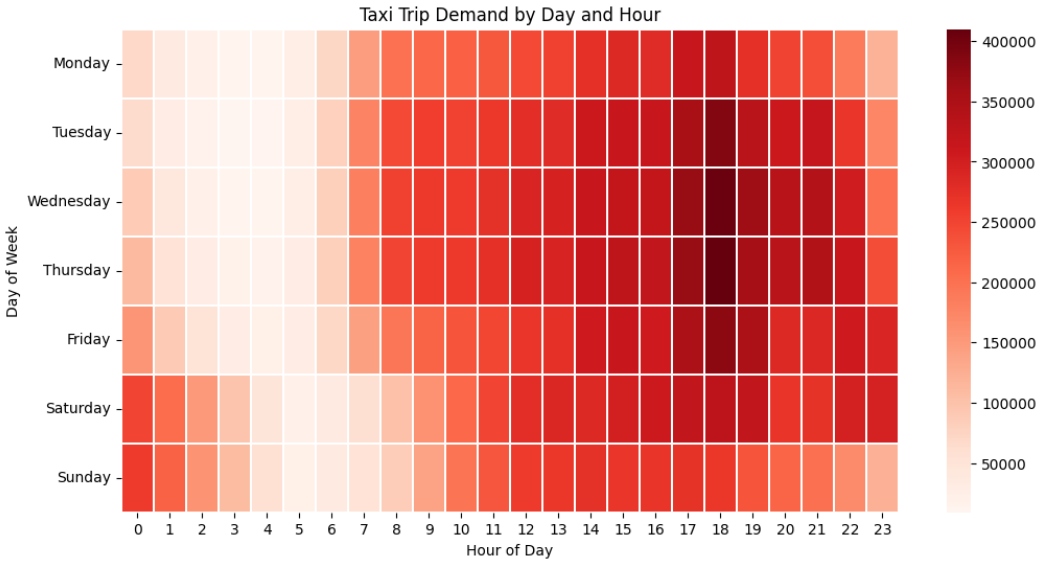


6 PM is the busiest hour, with Tuesday – Thursday having the highest taxi demand relative to other days at this time.

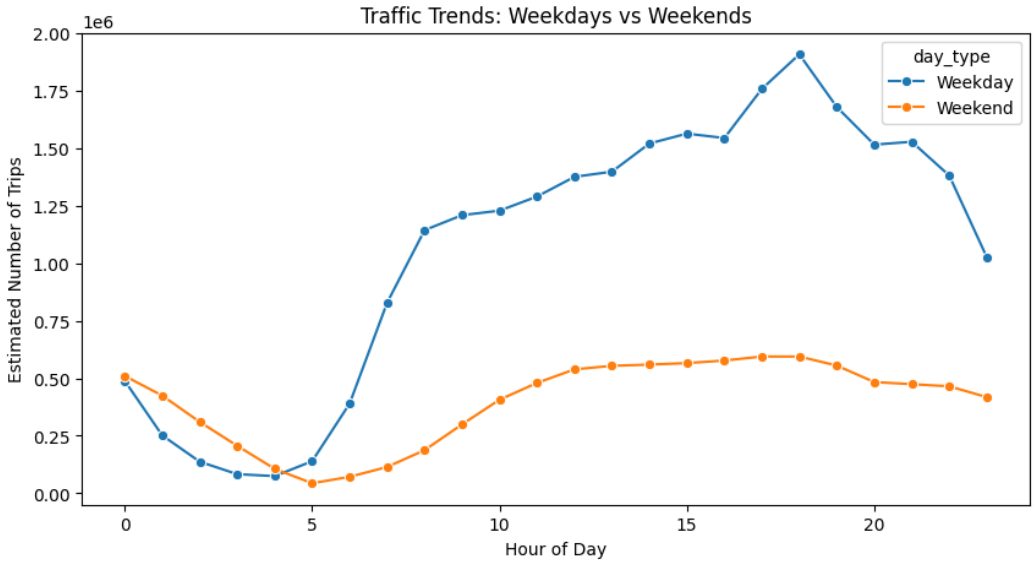
3.2.3. Scale up the number of trips from above to find the actual number of trips

To do the analysis, a sample of 5% was taken. Hence the actual number of trips would be ~20 times the sample.





3.2.4. Compare hourly traffic on weekdays and weekends



On a weekday, the taxi demand is very low during early morning hours (2 AM – 5 AM), and it rises sharply after 6 AM, indicating the start of morning commute traffic. Trips increase steadily through the day. The highest peak occurs around 5–6 PM, corresponding to the evening commute. After 7 PM, demand gradually declines but remains relatively high until late evening. The weekday taxi demand is largely driven by work-related commuting patterns.

On weekends, the taxi demand is higher than weekdays during late-night and early-morning hours (12 AM – 4 AM). The drop in demand during early morning is less extreme than weekdays. Demand increases gradually from late morning onwards. The peak occurs later in the evening, but overall demand is lower than weekday peaks. Weekend taxi usage is influenced more by leisure activities, nightlife, and social travel rather than commuting.

How Identifying Busy and Quiet Hours Helps

1. Better Fleet Allocation

Taxi companies and ride-hailing services can position more vehicles during busy hours to meet demand and reduce passenger wait times.

2. Reduced Idle Time for Drivers

During quiet hours, drivers can reposition to high-demand areas and reduce unnecessary cruising thereby increasing operational efficiency.

3. Dynamic Pricing Strategies

Ride-hailing platforms can use busy-hour insights to implement surge pricing and manage supply-demand balance.

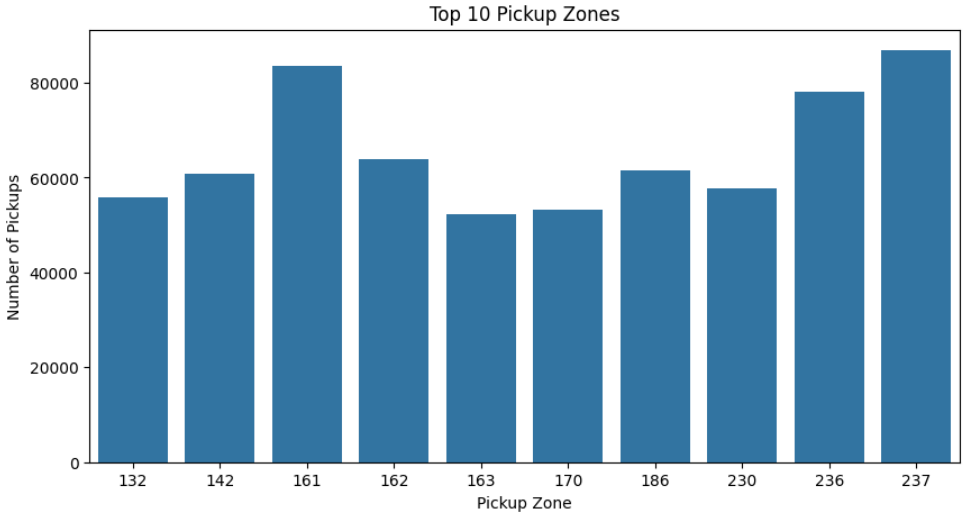
4. Traffic and Infrastructure Planning

City planners can use these patterns to understand peak congestion periods, improve traffic management strategies and optimize public transportation schedules.

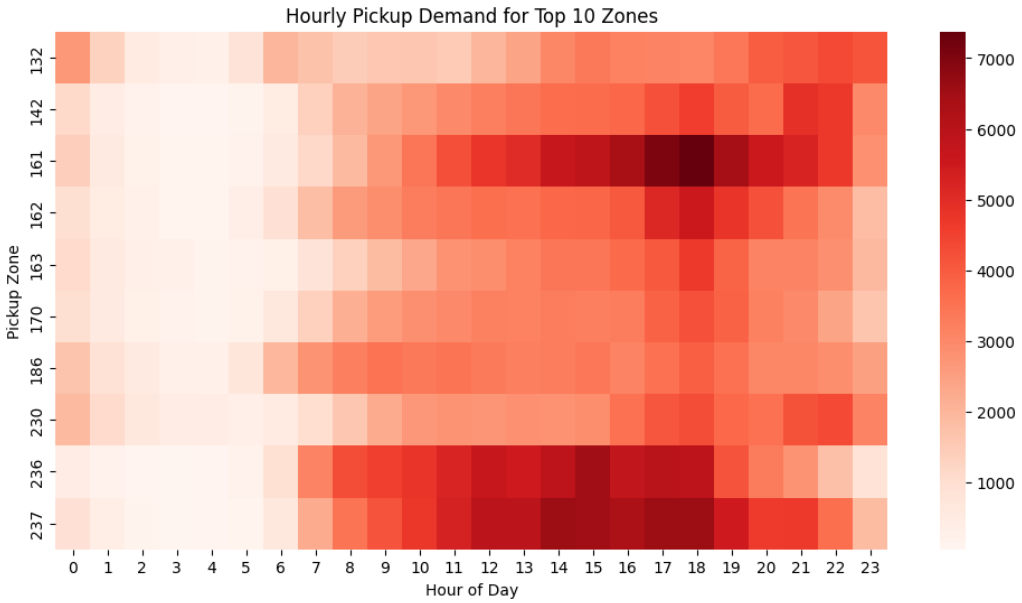
5. Better Demand Forecasting

Understanding these patterns enables more accurate predictive models for taxi demand, which helps companies prepare for high-demand periods.

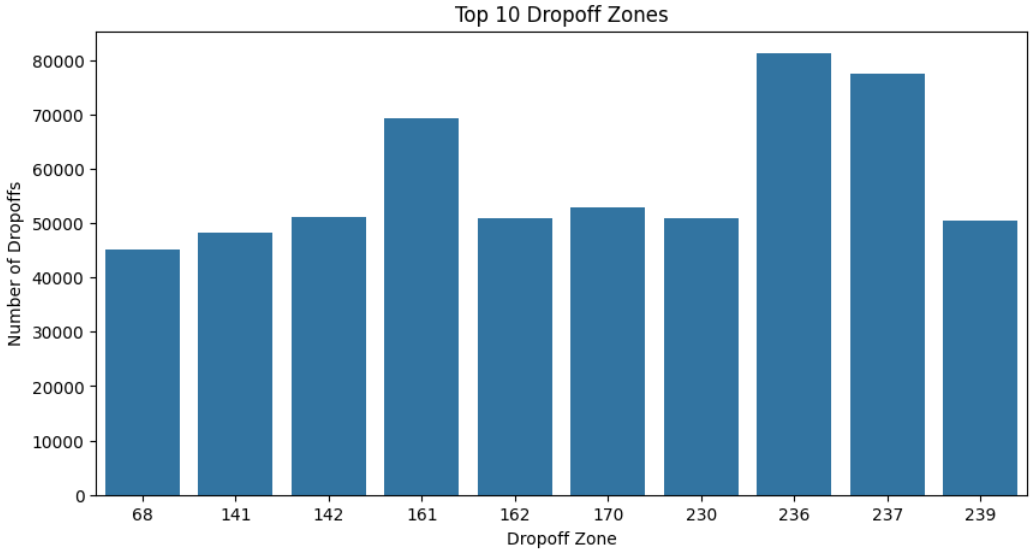
3.2.5. Identify the top 10 zones with high hourly pickups and drops



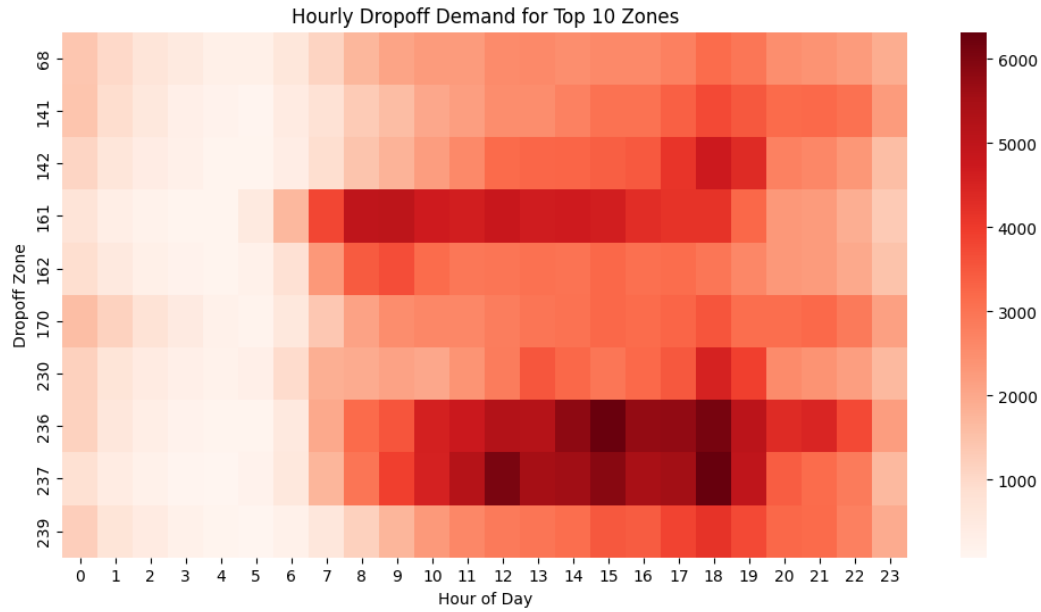
The zones 161, 237, and 236 has the highest number of pickups.



For the top 3 pickup zones, the number of pickups are heavily concentrated at the daytime office hours (8 AM – 7 PM). The number of pickups drop after 12 AM and does not rise up to 7AM.

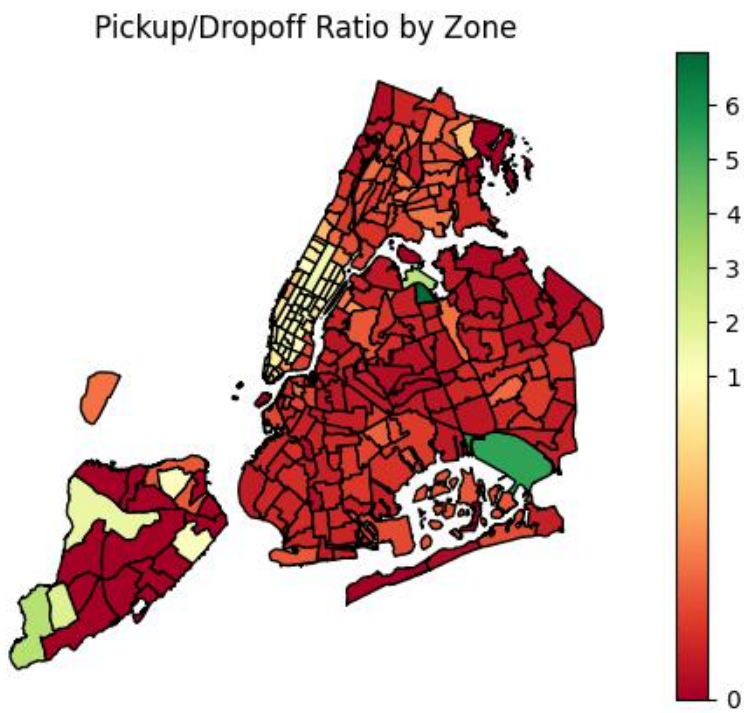


The zones 236, 237, and 161 has the highest number of dropoffs.



The dropoff hourly trend is like the hourly pickup trend where the number of dropoffs are concentrated during the office hours, and this reduces post 12 AM and does not pickup till 7 AM.

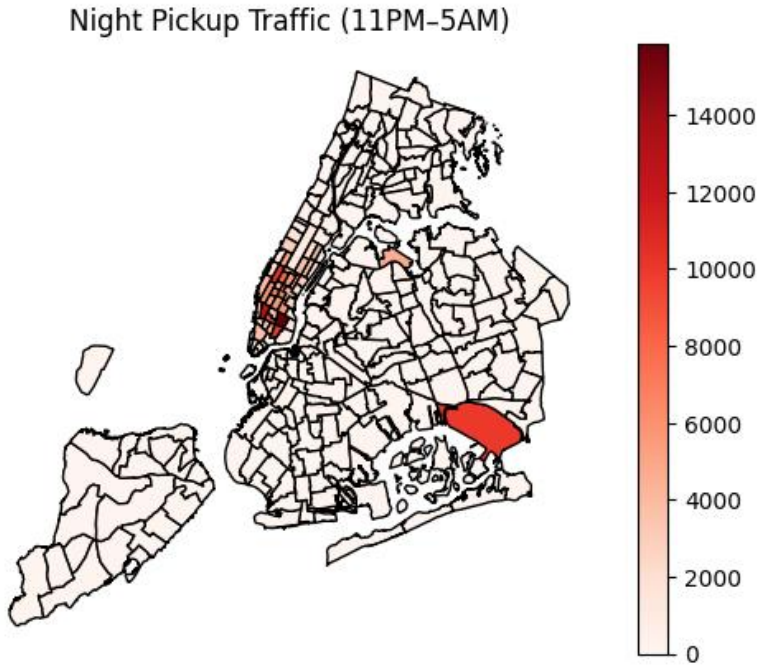
3.2.6. Find the ratio of pickups and dropoffs in each zone

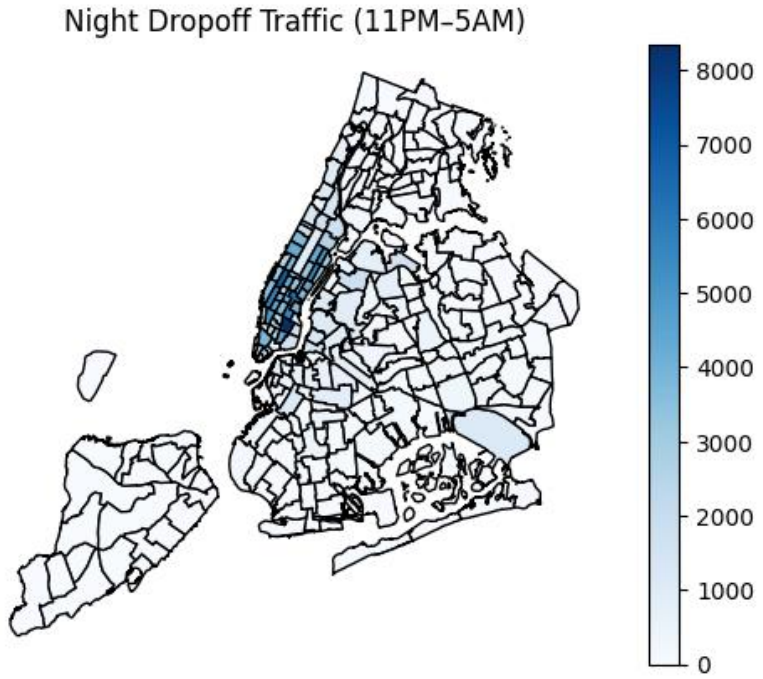


For the above visualization, zones in red have more dropoffs than pickups, zones in yellow have approximately balanced traffic and the zones in green have more pickups than dropoffs. The colour scale is modified to have the pickup-dropoff ratio at 1 as the center so that the

visualization can be interpreted easier. This choropleth map reveals that taxi traffic in NYC is highly directional, with Manhattan functioning as the primary destination hub, while outer borough zones often serve as trip origins. The visualization effectively highlights the spatial imbalance between where trips begin and where they end.

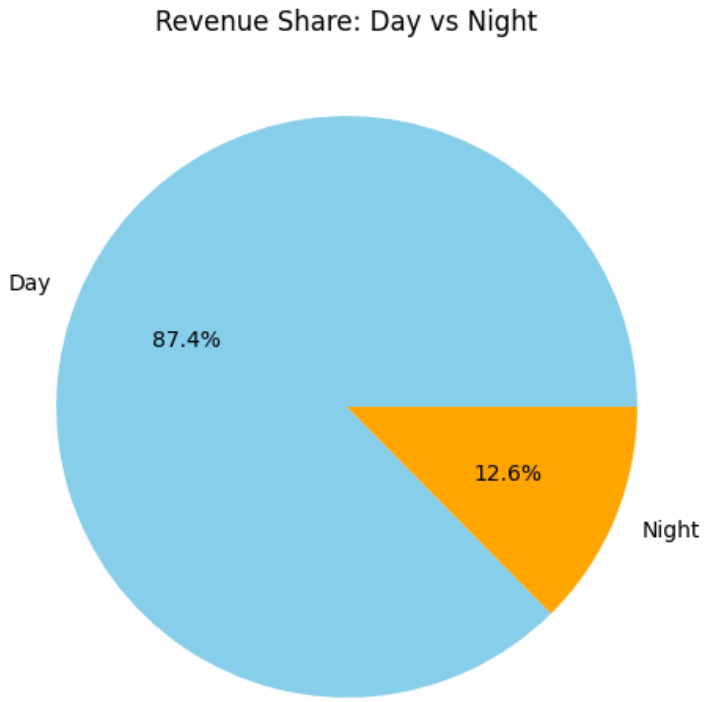
3.2.7. Identify the top zones with high traffic during night hours





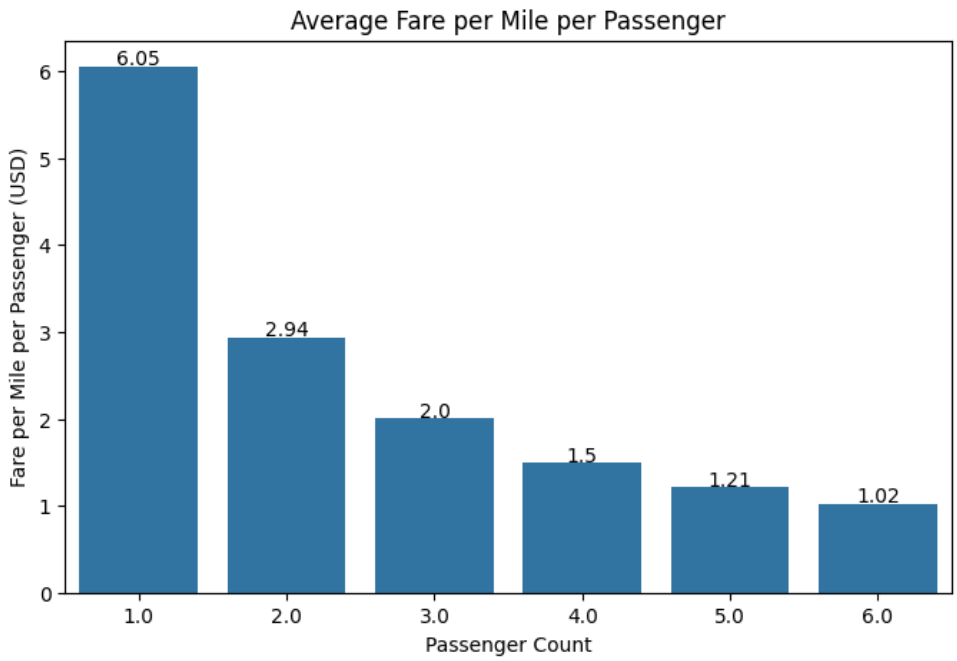
Both pickup and dropoff maps show the highest concentration of taxi traffic in Manhattan, particularly Midtown and Lower Manhattan. Airport zone shows significant night pickups. LaGuardia Airport shows noticeable pickup activity. Outer boroughs have lower night taxi activity. Nighttime taxi demand in NYC is largely driven by entertainment, tourism, and late-night travel, with Manhattan serving as the primary hub for both pickups and dropoffs, while airport zones also contribute significantly to late-night mobility patterns.

3.2.8. Find the revenue share for nighttime and daytime hours



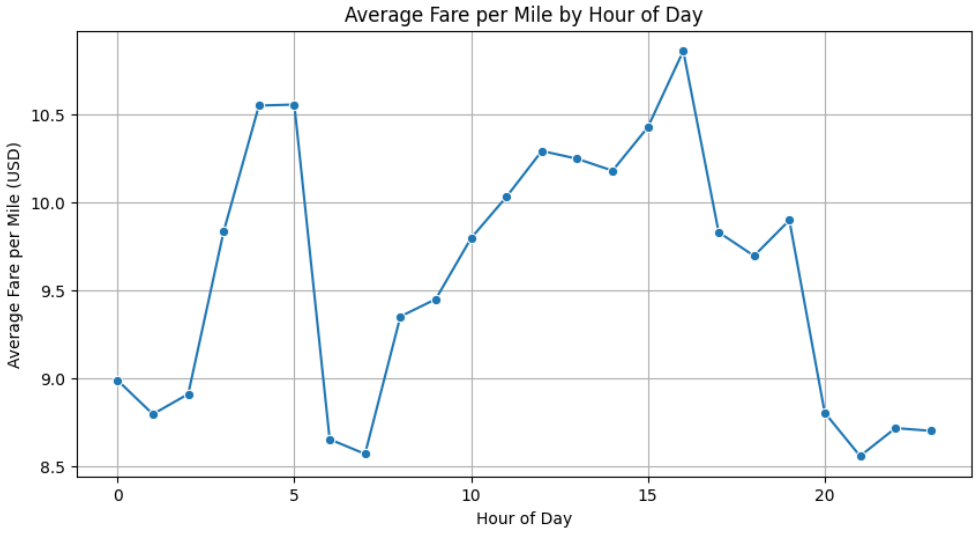
87.4% of total revenue is generated during daytime hours. Only 12.6% of total revenue is generated between 11 PM and 5 AM. This suggests that taxi demand is significantly lower during late-night hours.

3.2.9. For the different passenger counts, find the average fare per mile per passenger



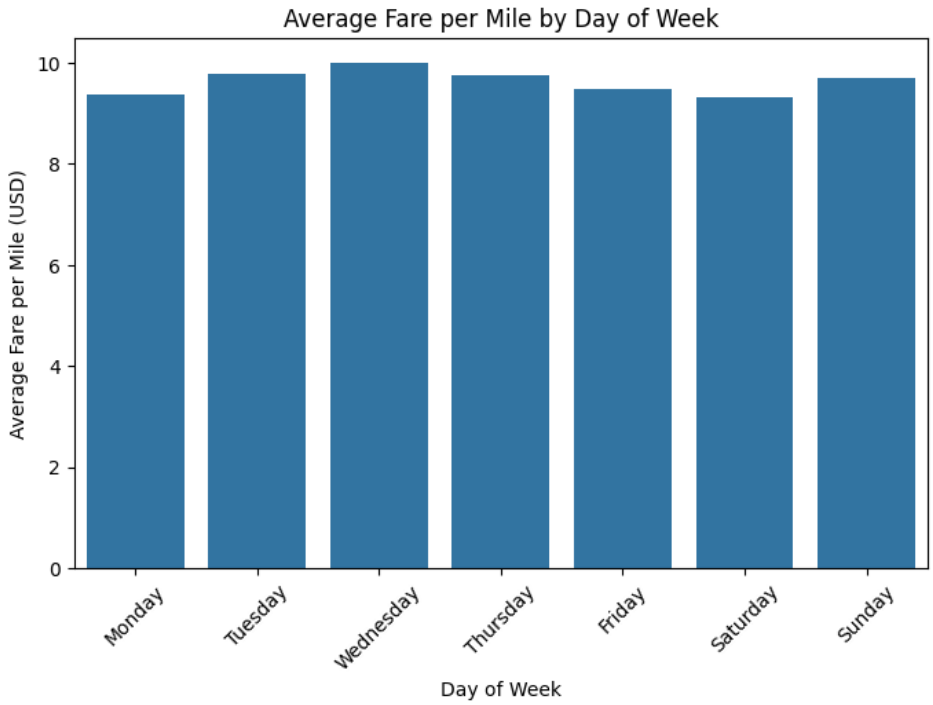
Cost per passenger decreases as passenger count increases. Solo riders pay the highest cost. Cost Reduces Significantly when passengers increase. Adding more passengers drastically reduces the individual cost per mile. Going from 1 to 2 passengers nearly halves the cost. Going from 2 to 4 passengers reduces the cost further by ~50%. The decrease becomes smaller after 4 passengers.

3.2.10. Find the average fare per mile by hours of the day and by days of the week

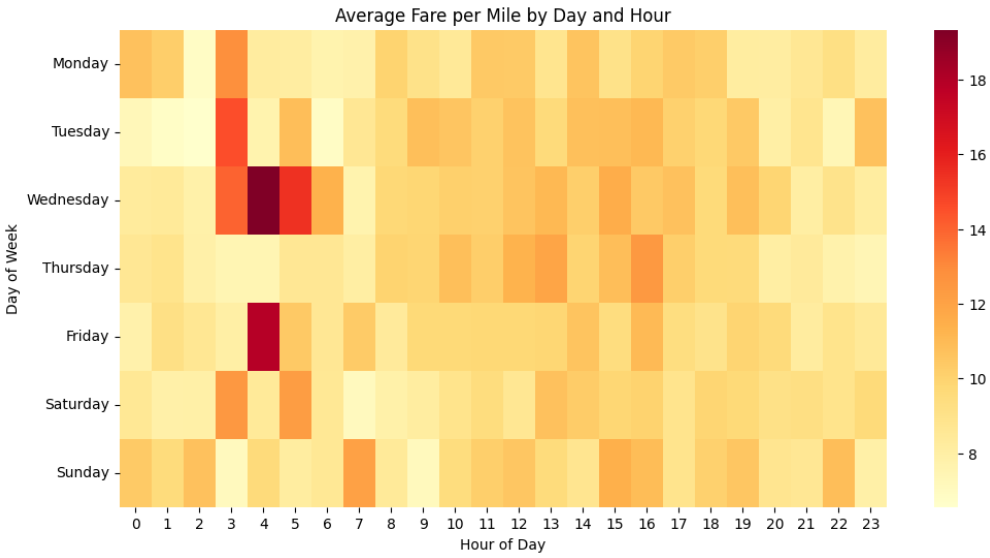


Late-night and early morning trips tend to generate higher fare per mile values. Fare per mile peaks around 3 AM – 5 AM, reaching above $10.5 per mile. The lowest values occur around 6 AM – 7 AM around morning commute. Another noticeable increase occurs around 3 PM – 4 PM,

reaching the highest values of fare per mile. Between 8 PM – 11 PM, fares per mile decrease.

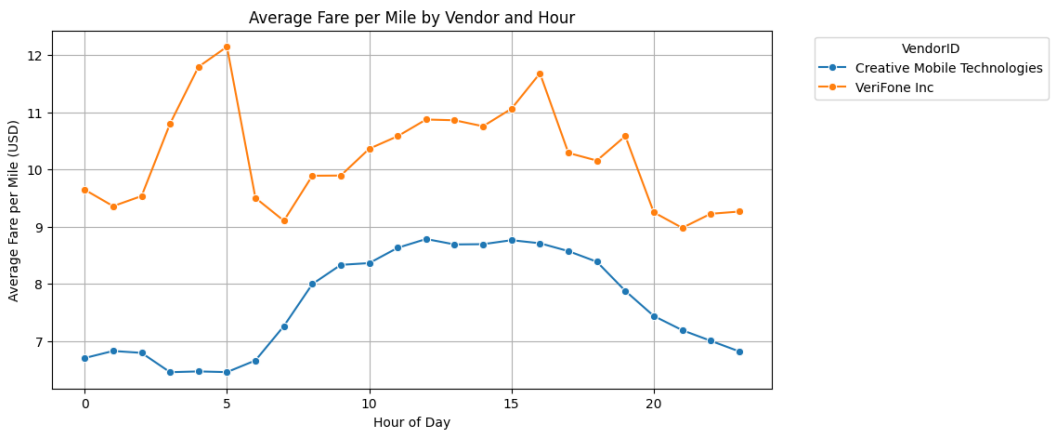


Fare per Mile is Relatively Stable Across Days. Average fare per mile ranges roughly between $9 and $10 per mile. Average fare per mile ranges roughly between $9.3 and $10 per mile. Overall variation between days is small, indicating that taxi pricing remains consistent throughout the week.



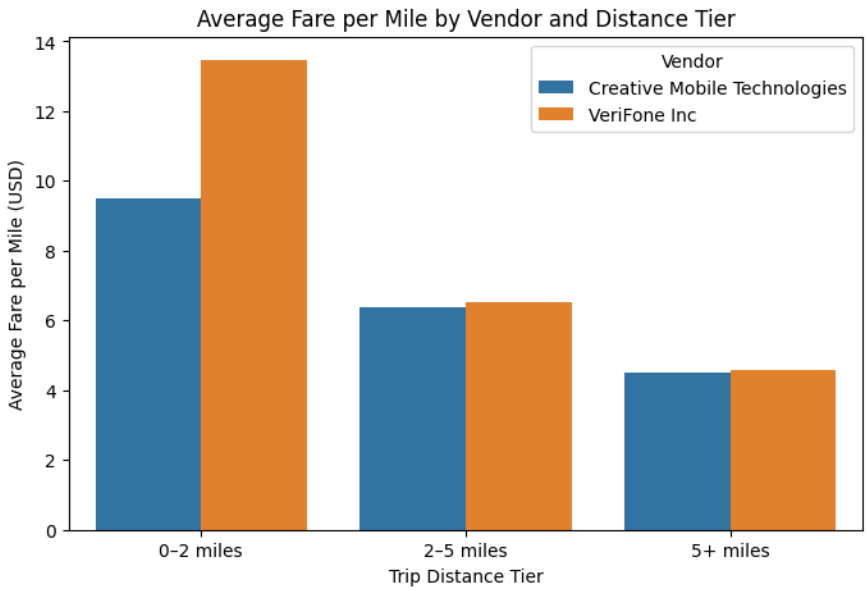
This heatmap shows how the average fare per mile varies across different hours of the day and days of the week. The darkest regions appear around 3 AM–5 AM, particularly on Wednesday and Friday. Between 6 AM and 9 AM, the heatmap shows lighter shades. This indicates lower fare per mile values. The hourly pattern is similar across most days of the week. Differences between weekdays and weekends are not very large. This indicates that time of day influences fares more strongly than the day of the week.

3.2.11. Analyse the average fare per mile for the different vendors



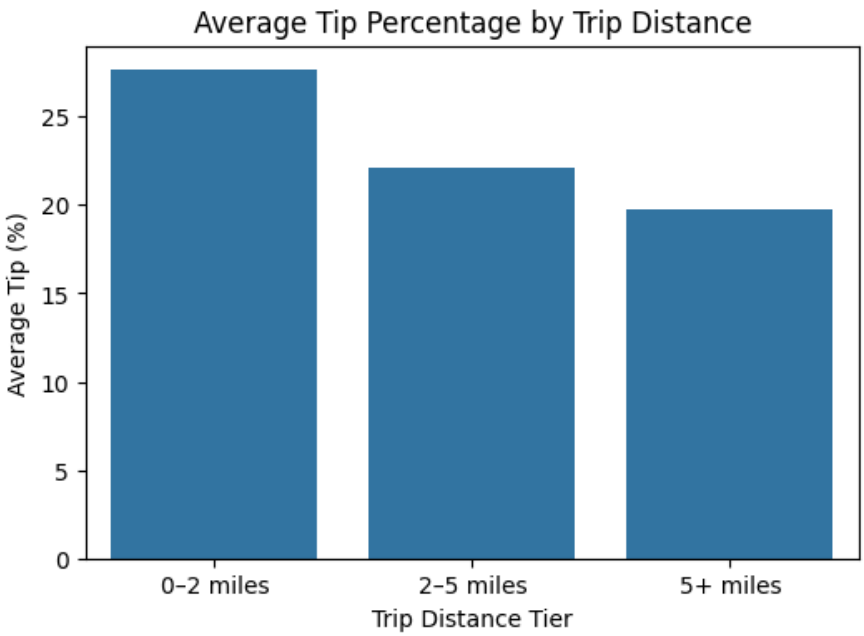
The average fare per mile charged by VeriFone is always higher than that charged by Creative Mobile Technologies for all hours of the day. Despite the difference in average fare per mile, both the vendors follow a similar trend throughout the day - higher fares per mile during late night and early morning hours, moderate fares during midday and gradual decline during late evening. This indicates that time-of-day travel patterns affect both vendors similarly.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

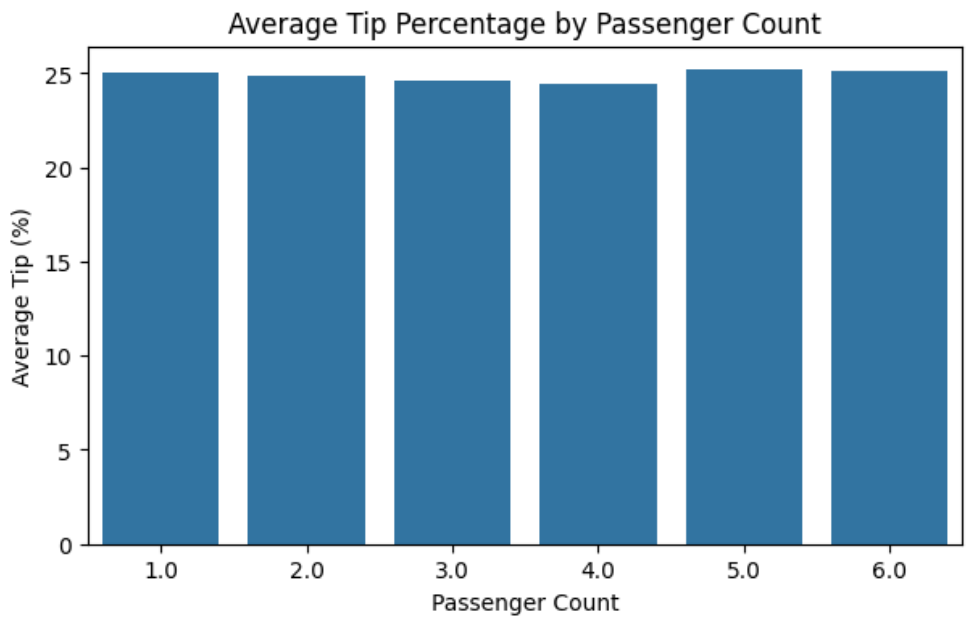


Short trips have the highest fare per mile, and it reduces with the increase in trip distance. For the trips in 0-2 miles segment, there is a significant increase in the average fare per mile charged by Verifone Inc than Creative Mobile Technologies. The average fare per mile is similar for both the vendors in other distance tiers.

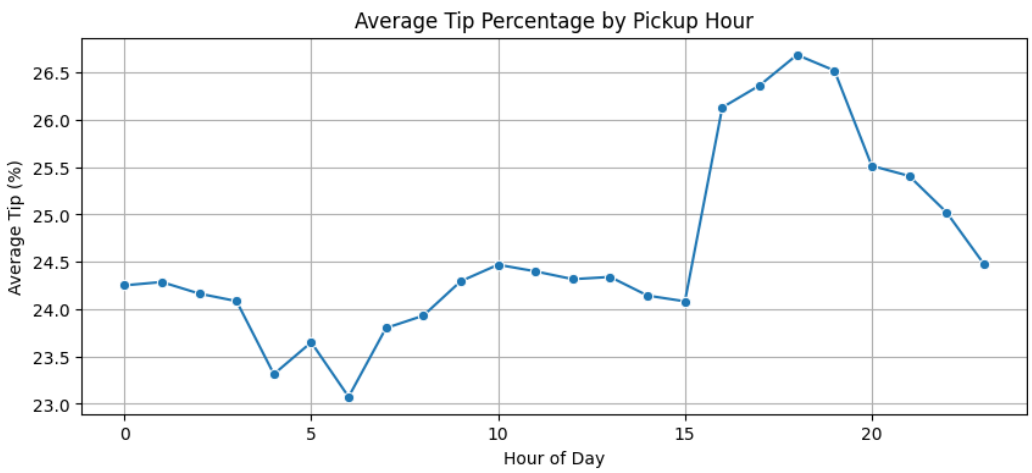
3.2.13. Analyse the tip percentages



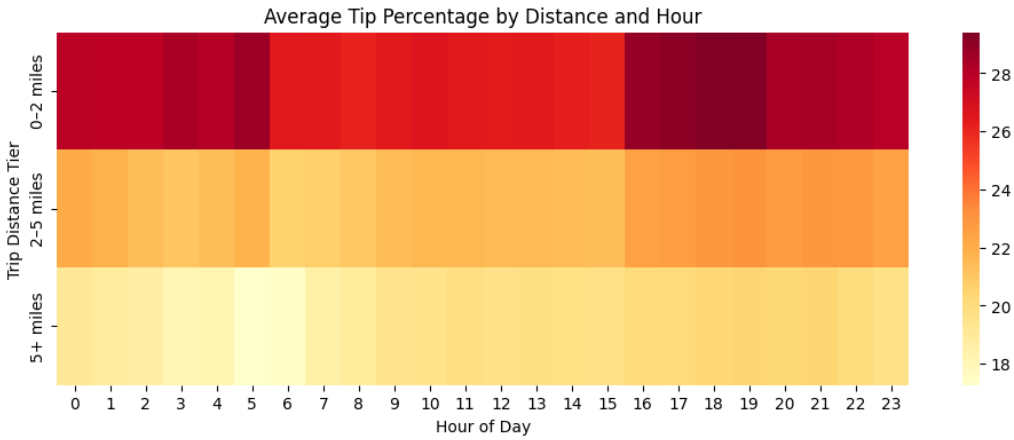
Tip percentage decreases as trip distance increases.



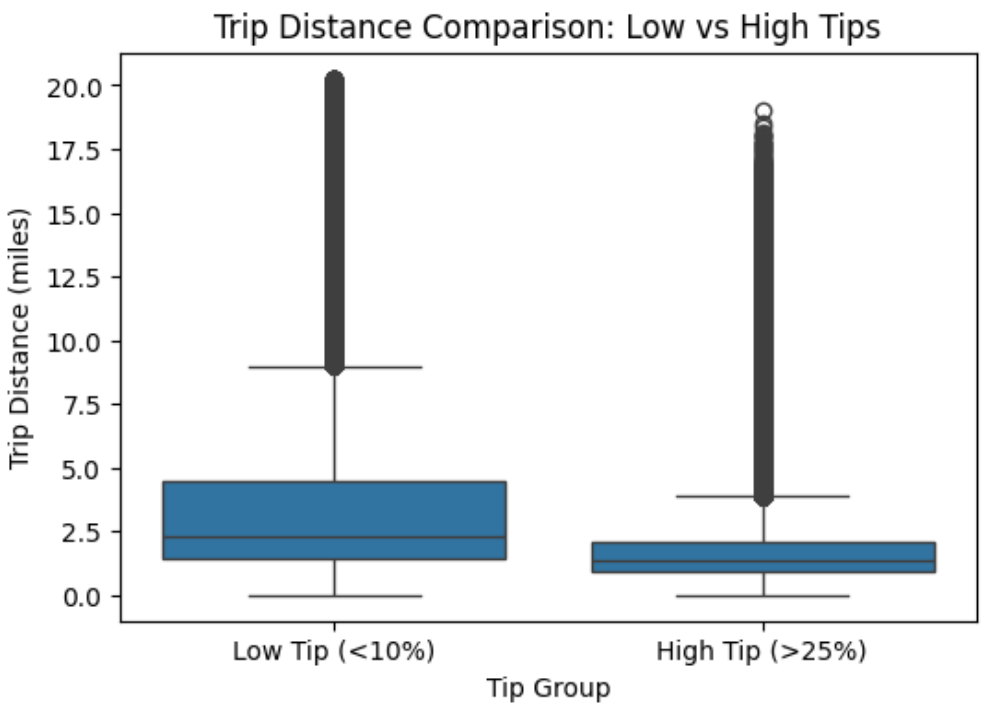
Tip percentages remain around 24–25% for all passenger counts (1–6). Hence, the number of passengers does not significantly influence tipping behavior.



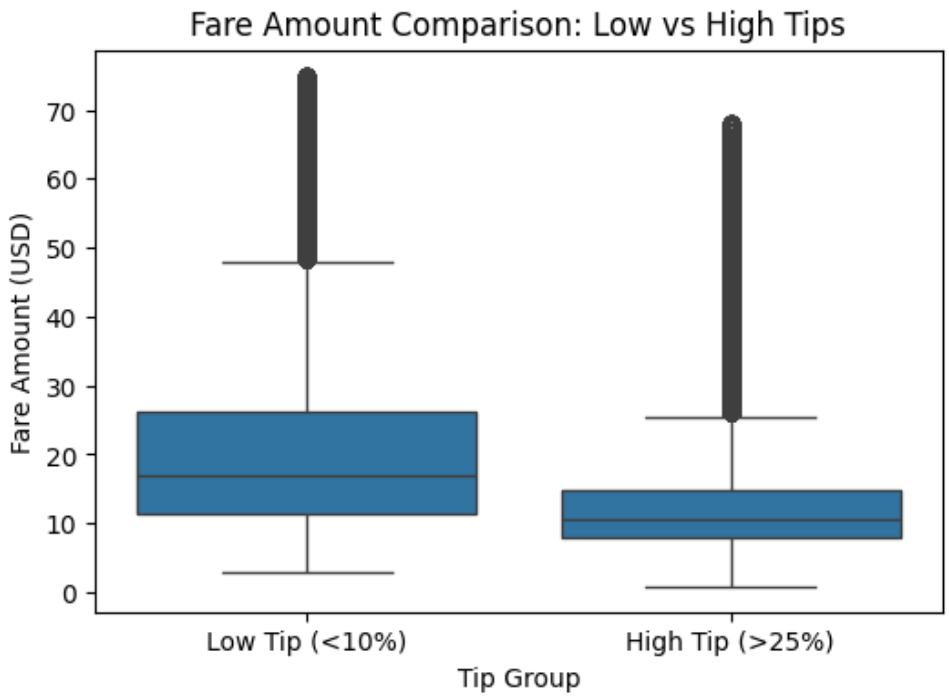
From the hourly tipping trend, we find that lowest tipping occurs around 4–7 AM. Tip percentages gradually increase through the day. Tipping is highest at around 4–8 PM.



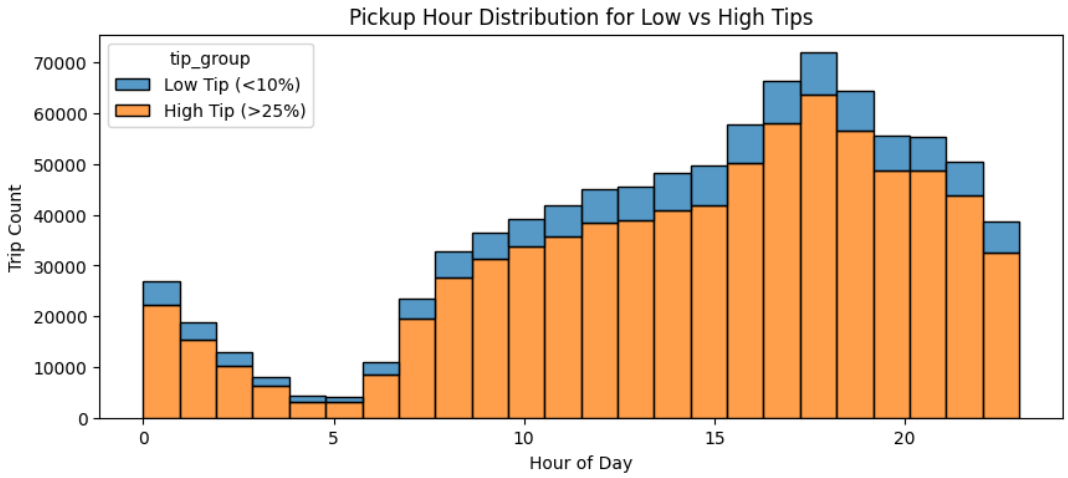
Short trips (0–2 miles) consistently show the highest tip percentages throughout the day. Long trips (5+ miles) show the lowest tipping percentages across all hours. Evening hours show slightly higher tipping for all distance tiers.



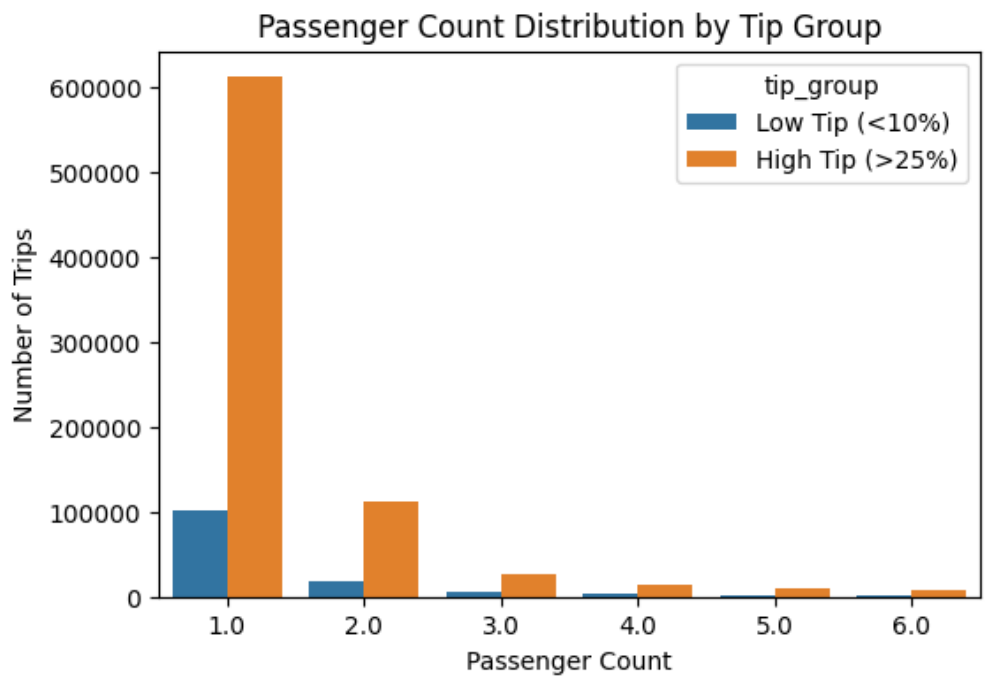
Trips with high tips tend to have shorter distances. Trips with low tips show a wider distribution and generally longer distances.



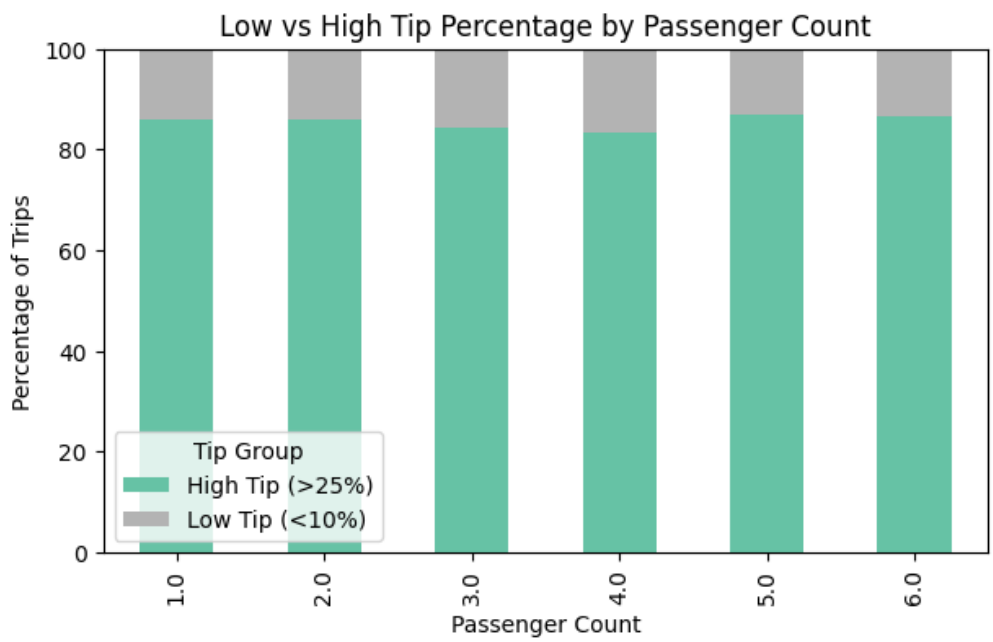
Lower fares are associated with higher tip percentages. High-tip trips have lower median fares. Low-tip trips tend to have higher fares.



Both low-tip and high-tip trips follow similar hourly patterns. Peak trip volumes occur during late afternoon and evening (around 5–7 PM) for both groups.

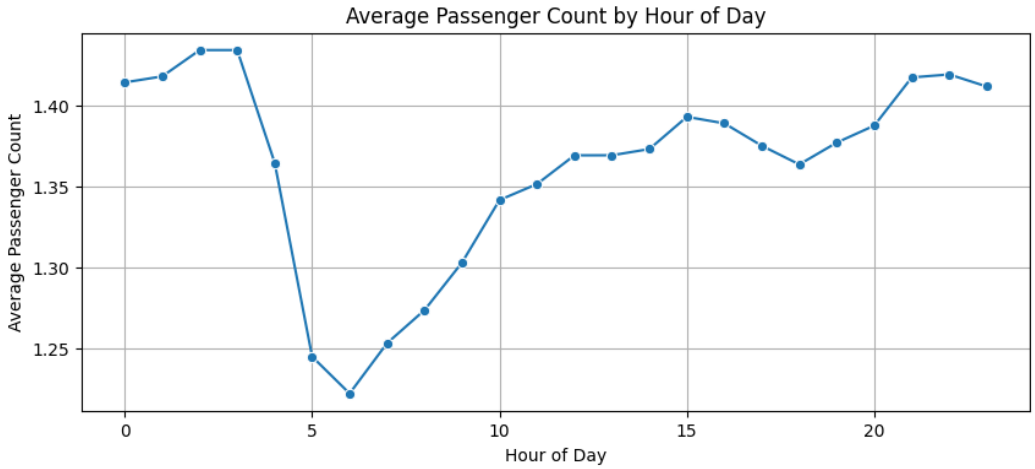


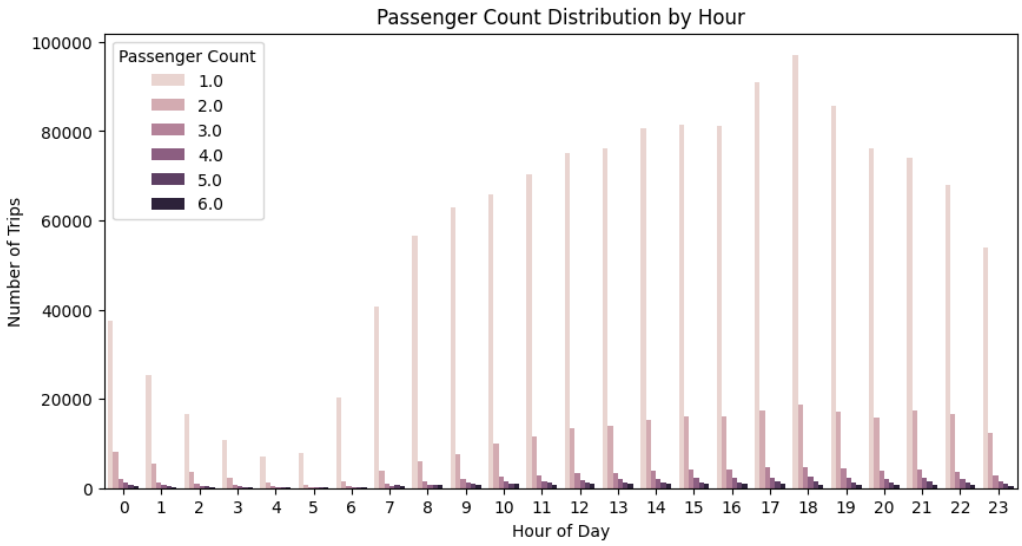
The majority of trips in both tip groups involve one passenger. Passenger counts decrease significantly for groups larger than two.



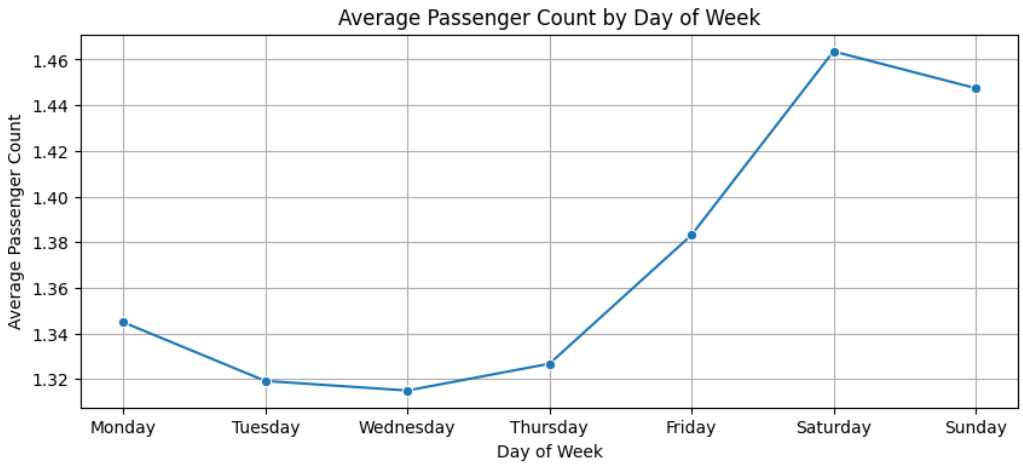
Passenger count does not significantly differentiate low and high tipping behavior.

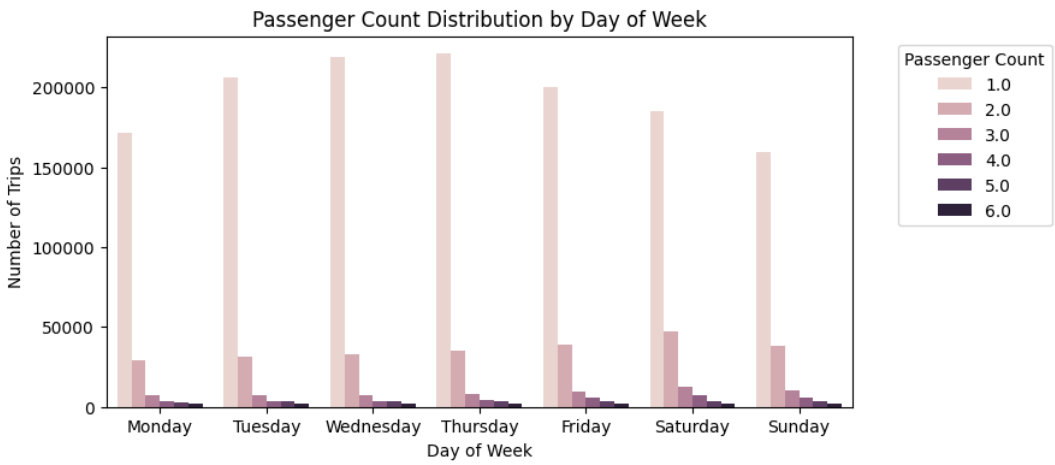
3.2.14. Analyse the trends in passenger count



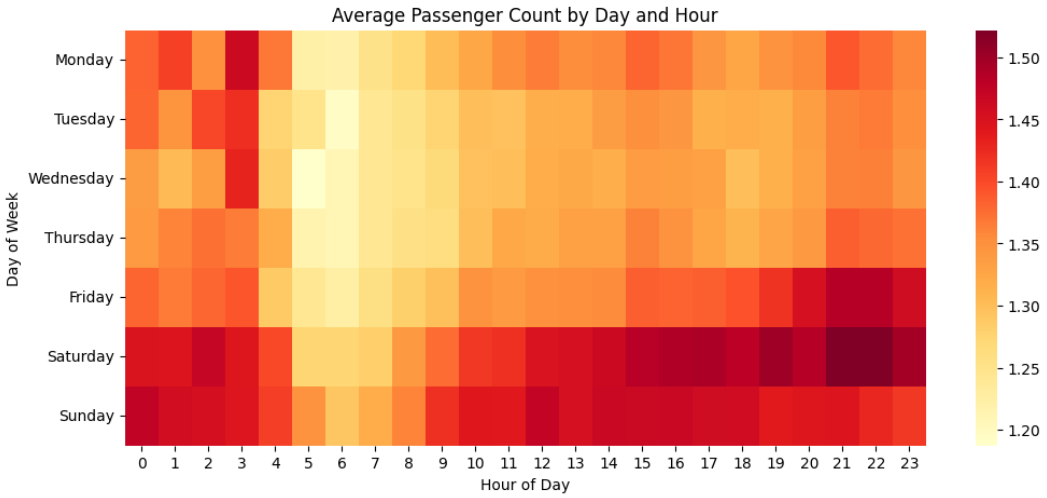


Trips with 1 passenger represent the majority of rides throughout the day, followed by 2. Trips with 3 or more passengers happen rarely. Lowest passenger counts occur during early morning hours. Passenger counts gradually increase during the day. Evenings show higher passenger counts.



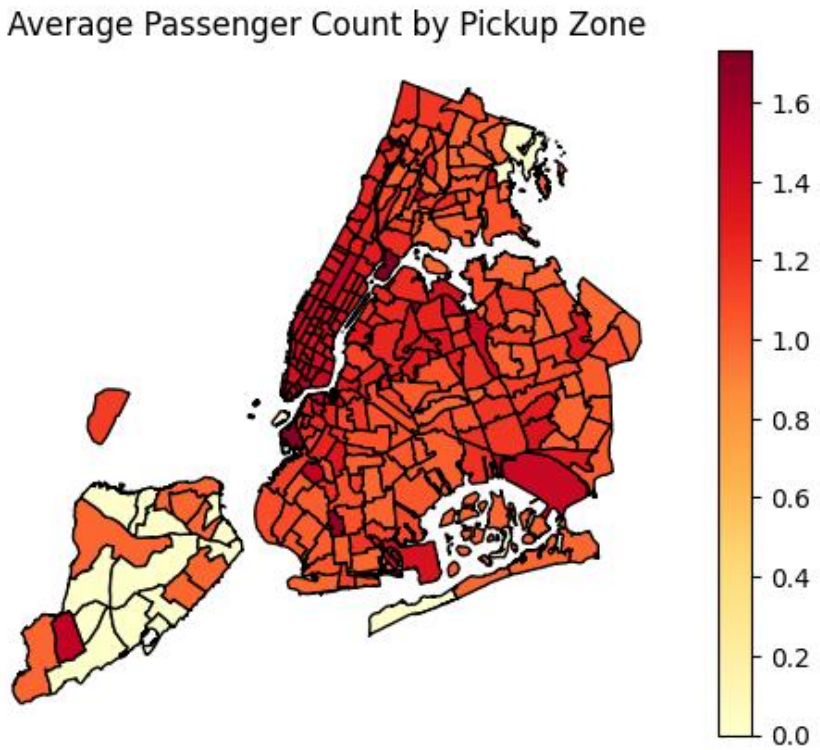


Weekends show the highest average passenger counts. Midweek has the lowest passenger counts. Passenger counts gradually increase toward the weekend. Single-passenger trips still dominate across all days.



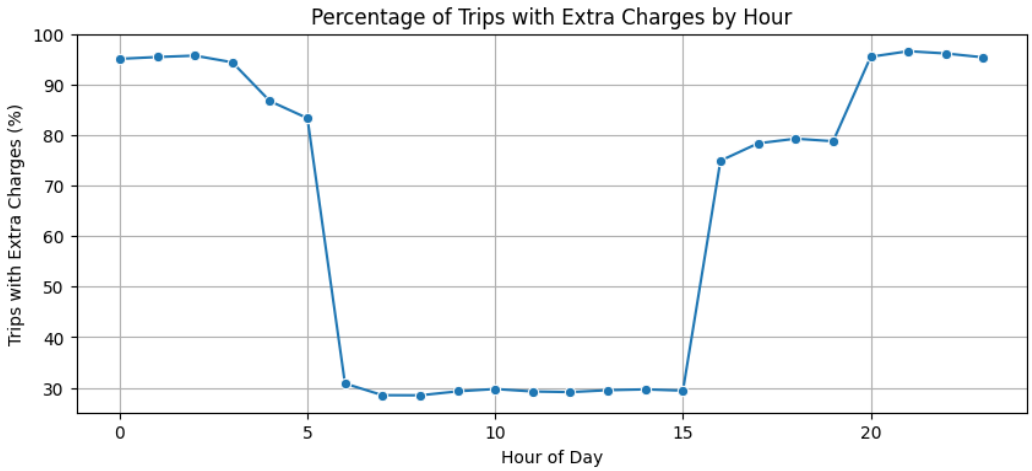
Passenger counts vary primarily based on time of day and day of the week. Weekday mornings are dominated by solo commuters, while evenings—especially on weekends—see more group travel. Despite these variations, the overall passenger count remains relatively low, indicating that most taxi trips involve one or two passengers.

3.2.15. Analyse the variation of passenger counts across zones

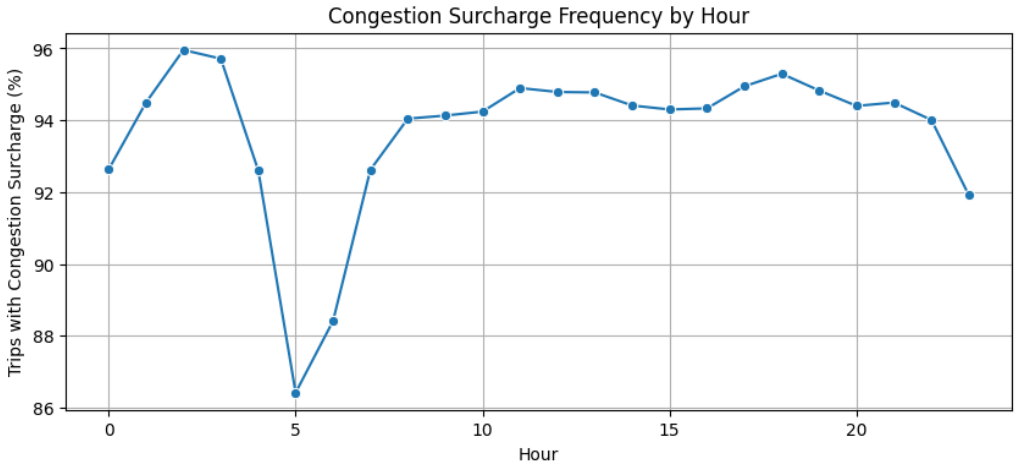


Most zones have low average passenger counts. Across most zones, the average passenger count lies roughly between 1.0 and 1.4 passengers per trip. The majority of taxi trips involve less number of passengers. Group travel is relatively uncommon. Manhattan shows slightly higher passenger counts. Several zones in Manhattan, particularly central areas, appear darker than surrounding regions. Zones around airports or major transport hubs tend to show slightly higher averages. Residential and outer borough zones show lower passenger counts. Passenger count variation across zones is relatively small. Even the darkest zones rarely exceed 1.6–1.7 passengers on average.

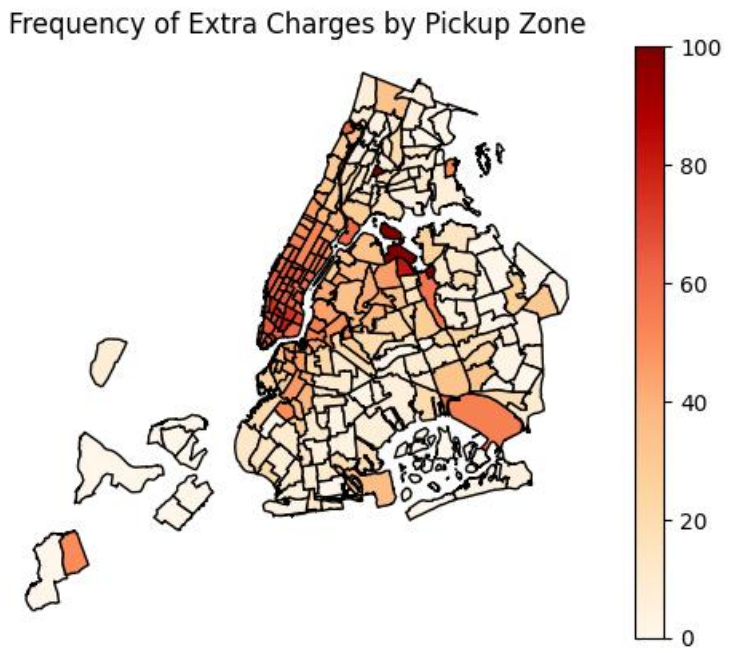
3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.



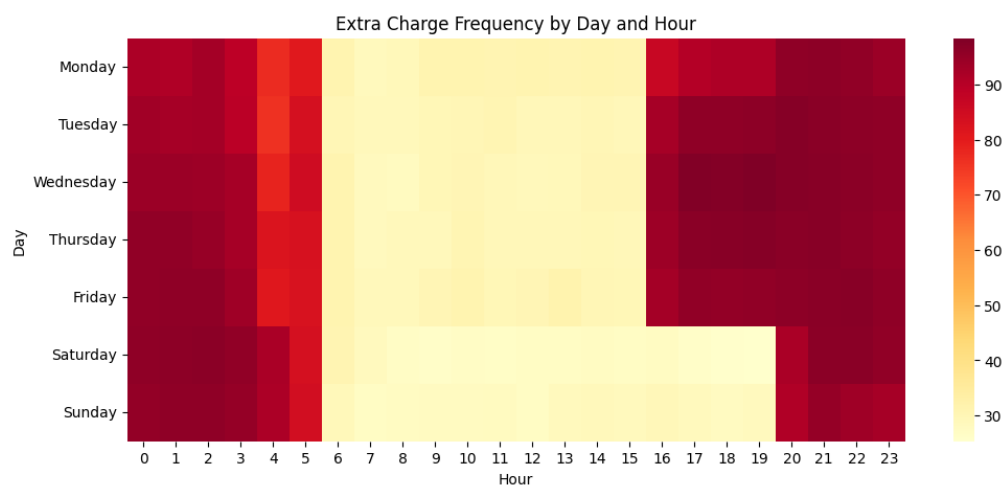
There are just around 30% of trips that charge extra charges for trips happening during 6 AM – 3 PM. Post 3 PM, this raises to around 75-80% till 7 PM. Post 7 PM more than 95% of the trips include extra charges.



From 8 AM to 10 PM, around 94% of the trips include congestion surcharge. This number is at the lowest at around 5 AM at around 86%. There isn't significant variation in the number of rides that include congestion surcharge during any hour of the day. This varies from 86 – 96% of rides at any given hour.



The trips in the Manhattan area and the areas surrounding the airport often include extra charges, which is highlighted in dark shades in the visualization. The trips in the outer boroughs less often include extra charges.



On the weekdays, the extra charges are more often added on trips happening post 4 PM up to 5 AM. On the weekends, extra charges are added to trips happening post 7 PM up to 5 AM.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

The busiest hours occur between 4 PM and 7 PM, peaking around 6 PM. Early morning hours (3 AM – 6 AM) experience the lowest demand. Demand increases sharply starting from 7 AM, corresponding to the morning commute. Increase fleet availability during evening peak hours, particularly 4–7 PM. Reduce idle capacity during early morning hours. Maintain moderate supply during morning commute hours (7–10 AM).

Thursday and Friday show the highest trip volumes among weekdays. Saturday evenings show the highest passenger counts, indicating leisure travel. Midweek days (Tuesday and Wednesday) show slightly lower activity. Increase fleet deployment during Friday evening and weekend nights. Slightly reduce fleet availability during midweek low-demand periods.

Tipping patterns reveal behavioral insights. Short trips receive the highest tip percentages (~27%). Tip percentages decline as trip distance increases. Evening hours (6–8 PM) show the highest tipping rates. Passenger count has minimal influence on tipping behavior. Evening shifts may generate higher driver earnings due to tipping. Incentivize drivers to operate during evening leisure travel hours.

Passenger count remains relatively stable. Average passenger count is around 1.3–1.4 per trip. Solo riders dominate taxi usage. Slight increases in passenger counts occur during weekends and evenings. Taxi services should primarily optimize for single-passenger rides. Larger vehicle fleets may be better utilized during weekend evenings.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

Manhattan zones dominate pickup activity, particularly Midtown. Airports and major transport hubs show higher passenger counts. Outer boroughs show lower pickup intensity. Position more taxis in Midtown Manhattan, Tourist districts and airport zones. Reduce idle taxis in low-demand residential areas during off-peak hours.

The pickup/drop-off ratio highlights supply-demand imbalances. Some zones have more pickups than drop-offs, indicating demand hubs. Other zones have more drop-offs, indicating passenger destinations. Rebalance

taxi supply by relocating vehicles from drop-heavy zones to pickup-heavy zones. Implement driver incentives to reposition vehicles to high-demand areas.

Night pickups are concentrated in entertainment districts and Downtown Manhattan. Drop-offs are more distributed across residential zones. Maintain taxi availability near nightlife hubs during late hours. Expect return trips to residential neighborhoods after midnight.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Extra charges occur more frequently in specific zones and time periods. Congestion surcharges occur mainly in Manhattan zones. Airport fees are associated with trips to JFK and LaGuardia airports. Night surcharges apply primarily during late evening hours. Drivers should be informed about high-surcharge zones where revenue potential is higher.

Revenue trends fluctuate slightly across the year. March – June generates the highest monthly revenues. Summer months (July–September) show slightly lower demand. Prepare for higher demand in spring and fall tourism seasons. Implement promotional pricing or incentives during summer dips.

Fare amount is strongly influenced by trip distance and duration. Fare increases with trip distance and trip duration. Short trips tend to have higher fare per mile due to base fare effects. Long trips have lower fare per mile but higher total revenue. Encourage drivers to serve short urban trips during high-demand periods. Maintain availability for longer trips to airports and outer boroughs.

Revenue can be optimized through dynamic pricing adjustments. Possible approaches include slight surge pricing during evening peak hours, lower fares during low-demand hours to stimulate demand and premium pricing for airport and high-demand zones. This maintains competitiveness while maximizing revenue.